Estimating employment transitions while accounting for classification error: evidence from South Africa*

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Abstract

Reliable estimates of employment transition rates are central to diagnosing the nature of unemployment and designing effective labour market policy. Distinguishing chronic from transitory unemployment hinges on accurate measurement of these churn rates. Yet measurement and classification error are pervasive in survey data, especially in lower income countries where well-targeted policies are most urgent and where alternative data sources, such as reinterview or administrative data, are scarce. Classification error in employment status is especially pernicious for transitions due to its well-documented multiplicative upward bias on churn estimates.

We develop a structural estimator that uses three consecutive waves of panel data to jointly identify true transition rates and misclassification

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probabilities, requiring no reinterview or administrative data. Under the assumption that the latent, true employment state is a first-order Markov process while the observed state may be misclassified, the distribution of reported employment statuses displays Markov violations that separately pin down transition and error parameters. We derive a modular framework, estimated using the expectation-maximization (EM) algorithm that allows for survey-relevant extensions. Applied to the South African Quarterly Labour Force Survey, we find a misclassification probability of approximately 3%, which causes conventional methods to overstate quarterly employment entry and exit rates by a factor of nearly three—from roughly 8% to roughly 3%. Despite small error rates, the multiplicative bias on flows is large, materially altering inferences about persistence versus churn and, hence, policy emphasis. These results are robust to extensions that incorporate (i) observed and unobserved heterogeneity using respondent covariates and a finite mixture over latent worker types, since heterogeneity can mimic Markov violations; and (ii) duration-dependence by directly relaxing the Markov assumption and allowing second-order dependence in the latent process. These extensions prevent over-assigning all Markov violations to misclassification when heterogeneity or duration are the true sources of a small number of transient, mean-reverting employment spells. Three additional findings reinforce the conclusion that classification error is present and materially inflates estimated churn and attenuates persistence. First, we exploit the fact that true duration progressions are deterministic conditional on the true employment histories in a tenure-contamination model to structurally identify transition rates by using information from the joint distribution of tenure and employment status. Tenure data suggested even higher employment misclassification rates than the baseline model, and find that employment churn is overestimated by a factor of 2. Second, tightening matching criteria across waves lowers both estimated churn and inferred misclassification, consistent with mismatches inflating estimated churn. Third, allowing misclassification to vary with cross-item reliability shows that interviews with inconsistent age progressions are more likely to display patterns consistent with misclassification, suggesting churn is due to noise in the survey rather than genuine mobility. Our results imply that South African employment is considerably more persistent than previously

characterised, with important implications for the design of labour market policy. The baseline estimator requires only three waves, no reinterviews or administrative linkages, making it scalable and cost-effective for survey programs.

Keywords: Misclassification error, employment transitions, EM algorithm, panel data, South Africa
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Contents

1 Introduction

Distinguishing chronic from transitory unemployment is central to diagnosing the nature of joblessness and designing effective labour market policy. In economies with high labour market churn—where workers are rapidly hired and fired—firms typically have market power. The policy challenge is to protect workers against frequent spells of unemployment, and policies favouring stronger worker protections and enforcement are appropriate (Aghion et al., 2006). By contrast, if employment is highly persistent, unemployment reflects a failure of job creation rather than a failure of retention; the policy prescription shifts toward lowering hiring costs and removing barriers to mobility (Nattrass, 2000). Because transition matrices are the primary tool for assessing the degree of labour market churn, credible measurement of entry and exit rates is not merely an academic exercise—it is a prerequisite for coherent policy design.

Labour force surveys are the workhorse for measuring employment transitions with transition matrices. Yet survey errors are ubiquitous, and classification error in employment status is especially damaging for transition estimates. The effect of misclassification on flows is multiplicative: a single transient error at wave t makes a chronically non-employed worker appear to have found a job and then lost it, fabricating two transitions that did not occur. Another misclassification in the opposite direction—a truly employed worker is observed as non-employed—does not cancel out the first error as when estimating levels; instead, it creates two additional spurious transitions, first an observed job loss followed by an observed job gain. Even low misclassification rates therefore produce severe upward bias in estimated flows (Porteba and Summers, 1986; Singh and Rao, 1995; Meyer, 1988). The same worker also appears to exhibit negative duration-dependence—a recent "entrant" who immediately "exits"—when in reality the worker never moved. Classification error thus simultaneously overstates churn and artificially attenuates persistence.

A large literature has documented the importance of classification error in employment surveys. The pioneering work of Porteba and Summers (1986), using reinterview data for the United States Current Population Survey (CPS), found that correcting for classification error reduced the monthly probability of job loss from 5% to 1.9% and job entry from 7.2% to 2.6%. Chua and Fuller (1987) found, using the same data, a 6% chance of misclassifying a truly unemployed individual as employed. Singh and Rao (1995) estimated misclassification probabilities of less than 1% in Canada, yet showed that even such small error rates produce substantial upward bias in transition estimates. Abowd and Zellner (1985) introduced the error-correction matrix approach that underlies much of this work. Reinterviews, however, are costly and rare in lower income countries (Ashenfelter et al., 1986), and themselves imperfect—respondents may change status between the original and re-interview survey, or modality differences between telephone and in-person interviews may introduce systematic bias (Markhof et al., 2026).

Motivated by these limitations, several estimators have been proposed that identify misclassification from panel data alone. Biemer and Bushery (2000) introduced Markov latent class analysis (MLCA) to the three-wave CPS data and showed that the resulting estimates closely match reinterview-based corrections. Their approach imposes a first-order Markov assumption on the true process and a local independence assumption on the errors. Feng and Hu (2013) use an eigenvalue decomposition of the response classification matrix to estimate misclassification rates and correct unemployment stock estimates for the United States, finding that official unemployment is understated by approximately 2.1 percentage points. Shibata (2019) develops a hidden Markov model (HMM) formulation that accommodates individual-level heterogeneity in the latent process, returning to maximum likelihood estimation rather than the eigenvalue approach.

We build on this tradition and develops a structural estimator that jointly identifies true employment transition rates and misclassification probabilities using only three consecutive waves of panel data and no reinterview or administrative validation. The baseline estimator is based on an Expectation–Maximisation (EM) algorithm that exploits a simple and powerful identification insight: if the latent true employment process follows a first-order Markov chain but observed employment is measured with error, then the observed three-wave distribution will systematically violate the Markov property in a way that separately pins down the transition rates and the misclassification probability. The EM algorithm converts this insight into a sequence of closed-form updates, making estimation efficient, numerically reliable, and easily extended. Our estimator builds on existing structural approaches in some important and novel ways.

We derive and implement a modular EM algorithm that yields closed-form M-step updates for the baseline model, making estimation scalable to large samples while allowing straightforward extension. We also develop a generalized EM (GEM) extension that allows observable worker heterogeneity via probit transition probabilities and a finite mixture model (FMM) for unobservable heterogeneity. Both of these, observed and unobserved heterogeneity, can mimic Markov violations—a worker who always has high exit probability looks like a Markov violation if heterogeneity is ignored—and the GEM extension ensures these alternatives cannot masquerade as misclassification. This framework allows us to relax the Markov assumption on true, latent employment transitions by implementing a second-order autoregressive, AR(2), specification and using four waves of panel data. This paper is the first to do so. A common critique of the existing models is that employment is duration-dependent, and that the Markov assumption mistakenly attributes short employment spells for misclassification. The AR(2) model allows for the direct testing and controlling for duration dependence in the latent process.

We further extend the model into a novel tenure-contamination model in which job tenure provides support for the structural identification of classification error. This specification exploits the fact that true tenure progressions are deterministic conditional on the joint, latent distribution of employment across three waves and tenure in the first wave. This extension is based on the following intuition. Consider a worker who is truly employed in all three waves, who has a true employment sequence is E–E–E. Tenure should deterministically increase by 0.25 years. If the worker is misclassified as non-employed at wave 2, they will not report tenure. However, tenure in wave 3 will still be equal to tenure in wave 1 plus 0.5 years. This tenure progression is very different to a worker for whom the job loss was real, with a true employment sequence of E–N–E. In this case, true tenure in wave 3 will equal 0.25 years, since the worker was truly non-employed in wave 2 and tenure resets to zero. Observed tenure is also affected by measurement error, which we model as a contamination process in the style of Goldstein (1982) and Horowitz and Manski (1995), allowing us to separate tenure contamination from state misclassification and provide corroboration of the baseline result.

We apply the estimator to the South African Quarterly Labour Force Survey (QLFS), a rotating panel survey collected quarterly since 2008. South Africa provides an ideal empirical setting for two reasons. First, it has persistently high unemployment rates—around 30%—alongside a vigorous academic debate about the nature of joblessness, with important policy implications for the design of labour market interventions (Banerjee et al., 2008; Nattrass, 2000; Burger and Von Fintel, 2014). Second, there is well-documented evidence of measurement problems in the QLFS (Burger, 2016; Lechtenfeld and Zoch, 2014), and reinterview data are unavailable, making a structural approach attractive.

Using the QLFS data, we document a striking empirical puzzle that arises from the use of transition matrices, the standard approach for estimating employment entry and exit rates. Three-month employment entry and exit rates are between 8% and 9%, implying six-month rates of roughly 15% under the Markov assumption, which is implicit in the interpretation of transition matrices. But the observed six-month transition rates—computed directly from four-wave panels—are only about 10%, roughly one-third lower than the implied value. This divergence is inconsistent with a first-order Markov process on the observed employment status, but is exactly what one would expect if the latent true process satisfies the Markov assumption and observed employment is measured with transient error.

Our baseline EM estimator recovers a misclassification probability of approximately 3%, which implies true quarterly entry and exit rates of approximately 3%—a near-threefold correction relative to the naïve transition matrix estimates. The symmetric and asymmetric misclassification specifications produce virtually identical results, as do the stationarity-restricted and free-initial-employment-probability versions. The estimated misclassification probability remains stable—mostly between 2% and 3%—across all extensions controlling for observed heterogeneity (age, education, gender, race, contract type), unobserved heterogeneity (two-type finite mixture), and second-order Markov dynamics. When we allow the misclassification probability to vary with survey response reliability—proxied by inconsistencies in reported age progressions or education levels—we find that the baseline misclassification rate is approximately 2.2% for reliable respondents but rises to over 17% for those whose age changes by an implausible amount between waves. This mechanism evidence reinforces the interpretation that the estimated misclassification reflects genuine survey error, not a modelling artefact. Stricter wave-to-wave matching criteria, which reduce the likelihood of including mismatched observations, consistently lower both the measured transition rates and the estimated misclassification probability, providing a further design-based robustness check.

The tenure-contamination model reinforces the same interpretation using a different source of variation. It estimates that roughly one quarter of tenure reports are contaminated, while about three quarters remain clock-consistent; conditional on this measurement process, employment-status misclassification is again large enough to matter, with estimates between 8.6% and 10.0% across the tenure specifications. These estimates imply long employment and non-employment spells—roughly five to six years on average—and transition estimates around half of what is estimated from the raw transition matrices. Therefore, these findings corroborate the central conclusion from the transition-only models: much of the apparent short-run churn in the raw QLFS panel reflects measurement error rather than genuine rapid movement between labour market states.

The findings have substantive implications for the characterisation of South African unemployment. If true quarterly entry and exit rates are roughly 3%, the average employment spell is on the order of several years, not several quarters. Employment in South Africa is considerably more persistent than previously characterised by studies using raw transition matrices (Banerjee et al., 2008; Cichello et al., 2014; Kerr, 2018). This is more consistent with a labour market in which unemployment reflects scarce job creation rather than high churn, pointing toward supply-side and job-creation policies rather than policies targeted at worker protection or flexible hiring.

The remainder of the paper is organised as follows. Section 2 describes the QLFS data and documents the empirical puzzle motivating the approach. Section 3 presents the EM estimator and its extensions. Section 4 reports the main results. Section 5 reports the alternative specifications. Section 6 concludes. Full technical derivations for the baseline, AR(2), and tenure contamination models are provided in Appendices A, B, and C.

2 Data

2.1 The Quarterly Labour Force Survey

The Quarterly Labour Force Survey (QLFS) is a household-based sample survey conducted by Statistics South Africa, introduced in 2008 to replace the biannual Labour Force Survey. The survey is representative of the civilian non-institutionalised population of South Africa and collects information on employment status, earnings, industry, occupation, and individual and household characteristics. It has a rotating panel structure that allows a subset of individuals to be followed longitudinally.

We use the QLFS panel for the period 2010Q1 to 2019Q4. Individuals are matched across consecutive waves using their household and person identifiers. Statistics South Africa undertakes some consistency cleaning of the data prior to release (Statistics South Africa, 2013), but there is well-documented evidence of remaining measurement problems in the matched panel (Burger, 2016; Lechtenfeld and Zoch, 2014). Our sample is restricted to individuals aged 18 to 55 who have non-missing employment status in all available waves. We impose no restriction on gender, race, or other demographics, to ensure the results are representative of the working-age population as a whole.

For the three-wave baseline analysis, we retain all individuals observed in three consecutive quarterly waves, yielding a sample of approximately 405,000 person-wave triads. For the four-wave AR(2) analysis in Section 5, we additionally require a valid fourth wave, yielding approximately 317,000 observations. Survey weights are taken as the geometric mean of the three (or four) wave-specific weights to produce a balanced cross-wave weight (Kalbfleisch and Lawless, 1985).

2.2 The empirical puzzle

Table 1 presents summary statistics for the three-wave sample. Panel A shows that the sample spans the full working-age range, with a mean age of approximately 34 years and a mean education level of roughly 10 years. The employment rate is stable across waves at approximately 48%, consistent with South Africa's persistently high non-employment rate.

Panel B documents the transition rate puzzle that motivates our approach. The observed three-month entry and exit rates—the probability of moving from non-employment to employment, and vice versa, between two consecutive quarterly waves—are approximately 8% in both directions. Under the first-order Markov assumption, these three-month rates imply six-month rates of approximately $1 - (1 - 0.08)^2 \approx 15\%$. But the six-month rates observed directly in the four-wave data—the probability of transitioning over a six-month window, using only the first and third waves—are only around 10%. The implied six-month rate exceeds the observed six-month rate by a factor of approximately 1.5. This pattern—where transition rates estimated from high-frequency panels imply far more churn than lower-frequency estimates—is exactly the signature of transient misclassification error (Meyer, 1988; Porteba and Summers, 1986). Under the first-order Markov assumption on the *true* process, a transient error at wave t creates a spurious entry and a spurious exit, causing the high-frequency (3-month) estimate to be inflated while the low-frequency (6-month) estimate is less affected because the error has reversed by the time the second wave is observed.

This divergence is also consistent with the broader pattern in the South African labour market literature, where high-frequency panels consistently yield higher estimated transition rates than lower-frequency panels (Banerjee et al., 2008; Cichello et al., 2014). Burger (2016) shows that the Markov assumption applied directly to observed employment in South Africa is rejected by the data, with classical transition estimates failing to fit the observed three-wave moments. Our structural estimator attributes this violation to misclassification error, while simultaneously testing the alternative explanations of worker heterogeneity and duration dependence.

2.3 Evidence of measurement error

Panel C of Table 1 provides direct evidence that the QLFS panel contains substantial measurement error, using two variables—age and education—for which errors are more easily detected than in employment status. An age inconsistency is defined as a wave-to-wave change in reported age that is not in $\{0, 1\}$ years; since quarterly surveys are conducted 0.25 years apart, age should remain constant or increase by one because responses are rounded to the nearest integer year. An education inconsistency is a wave-to-wave change that is negative or exceeds one year, since education should be non-decreasing and cannot advance by more than one level per quarter. These inconsistencies represent clear measurement error: biological age cannot decrease, and educational attainment cannot be removed.

We find that approximately 11.6% of individuals in our sample have at least one age inconsistency across any pair of consecutive waves, and 26.8% have at least one education inconsistency. Overall, nearly one in three panel observations (31.2%) contains an inconsistent response in at least one of these two verifiable variables. These figures, which are, if anything, an underestimate of overall measurement error since they exclude undetectable errors in age (e.g., a plausible but false age of 35 reported as 34), suggest that measurement error is widespread in the QLFS panel. Crucially, we exploit the link between response unreliability and employment misclassification in Section 4, where we show that survey respondents who produce age or education inconsistencies also exhibit employment dynamics that are consistent with significantly higher misclassification rates.

2.4 Tenure and duration data

In addition to employment status, the QLFS collects information on job tenure (how long the employed person has been with their current employer) and non-employment duration (how long since the non-employed person last worked). These duration variables are used in the tenure contamination model described in Section 3.2.5. They also provide independent, direct evidence of employment misclassification. Table 2 summarises three diagnostic moments from the linked panel that make this evidence visible in the raw data.

Panel A begins with the tenure clock itself. Job tenure evolves deterministically: a worker who remains employed with the same employer between two consecutive quarterly waves should report a tenure increase of exactly three months. Among all observed employed–employed continuation pairs in the sample, approximately 65% report precisely this expected increment. The remaining 35% report tenure changes that deviate from the three-month clock—including negative changes, zero changes, and changes that exceed a year—in a pattern consistent with a contamination process rather than Gaussian measurement noise. This two-component structure, in which most reports are exact and a non-trivial minority are drawn from a dispersed distribution, motivates the point-mass–plus–contamination measurement model that we formalise in Section 3.2.5.

Panel B shows that tenure data also reveal direct evidence that observed employment status changes are sometimes spurious. Consider the subset of individuals observed with the sequence employed–non-employed–employed ($(s_1, s_2, s_3) = (1, 0, 1)$), who appear to have lost a job and then found a new one. If the job loss at wave 2 was genuine and the worker was re-employed at wave 3, tenure at wave 3 should be short—no more than three months. Instead, for nearly 30% of these individuals, tenure at wave 3 is approximately equal to tenure at wave 1 plus six months, exactly the progression expected if the worker was continuously employed throughout and the non-employment report at wave 2 was a misclassification. The tenure flanks are thus informative about which $(1, 0, 1)$ sequences reflect genuine transitions and which reflect classification error.

Panel C reports a complementary pattern among within-panel "fresh starts"—individuals observed as non-employed at wave $t-1$ and employed at wave t. If these transitions were genuine, the worker would have started a new job within the past three months, and reported tenure should be short. Yet approximately 37% of these apparent new entrants report tenure exceeding twelve months. A genuine job entrant cannot have more than one year of tenure with the current employer; these observations are therefore direct evidence of either misclassification of employment status at wave $t-1$ (the worker was actually employed all along) or measurement error in reported tenure. Either way, they confirm that observed employment transitions contain a substantial spurious component that is corroborated by a variable—tenure—that the employment status question does not directly reference.

3 Methodology

3.1 Baseline model

Setup and notation

Let $h_{it} \in \{0,1\}$ denote the true employment status of individual i at quarter t, where $h_{it} = 1$ indicates employment. We assume that the true process follows a first-order, time-homogeneous Markov chain with transition probabilities

$$\theta_0 = \Pr(h_t = 1 \mid h_{t-1} = 0), \qquad \theta_1 = \Pr(h_t = 0 \mid h_{t-1} = 1),$$

where we suppress the individual subscript for clarity. The stationary (long-run) employment rate is $\alpha = \theta_0/(\theta_0 + \theta_1)$. We consider both a stationarity-restricted specification in which the initial probability $\Pr(h_1 = 1) = \alpha$ and an unrestricted specification in which $\Pr(h_1 = 1)$ is estimated freely.

Survey respondents report $s_{it} \in \{0,1\}$, their observed employment status, which may differ from h_{it} due to measurement error. We assume a symmetric, time-invariant misclassification structure:

$$\pi = \Pr(s_t = 1 \mid h_t = 0) = \Pr(s_t = 0 \mid h_t = 1),$$

so that employment and non-employment are equally likely to be misclassified, and errors are independent across waves conditional on the true state. The full technical derivation of the likelihood and its properties, including the identification analysis, is given in Appendix A.

Table 1: Descriptive statistics: QLFS rotating panel, 2010Q1–2019Q4

Statistic	**Value**
Panel A: Sample (3-wave linked triads)	
Observations	405,658
Age range	18–55
Mean age (wave 1)	33.8
Mean education (years, wave 1)	10.2
% female	51.2%
Employment rate: wave 1 (%)	48.2
Employment rate: wave 2 (%)	48.2
Employment rate: wave 3 (%)	48.5
Panel B: Observed transition rates	
3-month rates	
Entry rate: wave 1→2 (%)	8.33
Entry rate: wave 2→3 (%)	8.13
Exit rate: wave 1→2 (%)	8.91
Exit rate: wave 2→3 (%)	8.26
6-month rates	
Observed entry rate: wave 1→3 (%)	10.41
Observed exit rate: wave 1→3 (%)	10.59
Implied entry rate (2-step Markov, %)	15.78
Implied exit rate (2-step Markov, %)	16.43
Ratio: implied / observed entry	1.52
Ratio: implied / observed exit	1.55
Panel C: Data quality – inconsistent responses	
% with age inconsistency (any wave)	11.6
% with education inconsistency (any wave)	26.8
% with either inconsistency	31.2

Note: All rates are weighted using the geometric mean of wave-specific survey weights. 3-month transition rates are the mean of wave 1→2 and wave 2→3 conditional rates. 6-month rates use 4-wave linked observations (N = 317,437). Implied 6-month rates compound the 3-month rates under the Markov assumption: $1-(1-r)^2$. Age inconsistency: consecutive wave gap not 0 or 1 years. Education inconsistency: non-monotone or multi-category jump.

Table 2: Tenure-based evidence of spurious employment transitions

Diagnostic moment	**N**	**Weighted share (%)**
Panel A: Employed–employed continuation pairs		
Exact tenure clock: $g_t - g_{t-1} = 3$ months	343,919	64.7
Non-clock increment: $g_t - g_{t-1} \neq 3$ months	343,919	35.3
Large non-clock deviation: $\lvert g_t - g_{t-1} - 3 \rvert > 12$ months	343,919	17.7
Panel B: Employed–non-employed–employed sequences		
Tenure flank follows continuous-employment clock: $g_3 \approx g_1 + 6$ months	6,025	29.8
Wave-3 tenure consistent with a genuinely new job: $g_3 \leq 3$ months	6,025	19.2
Panel C: Apparent within-panel job starts		
Short reported tenure at start wave: $g_t \leq 3$ months	35,618	38.6
Impossible long tenure at start wave: $g_t > 12$ months	35,618	37.4

Note: Rows report weighted shares among observations with the status pattern shown in each panel. N is the unweighted denominator: adjacent pairs in Panels A and C, and three-wave triads in Panel B. Tenure g is measured in months. For $(1, 0, 1)$ sequences, the continuous-employment clock is defined as $\lvert (g_3 - g_1) - 6 \rvert \leq 2$ months. Apparent starts pool observed $0 \rightarrow 1$ transitions over waves 1→2 and 2→3. All shares use the geometric mean of wave-specific survey weights.

Observed histories and marginal likelihood

For each individual observed in three consecutive waves, the observed employment history is the triple $(s_1, s_2, s_3) \in \{0,1\}^3$, yielding eight possible observed histories $\mathcal{H} = \{000, 001, \ldots, 111\}$. The marginal likelihood for an individual with observed history $\mathbf{s}$ is obtained by summing the joint likelihood of $(\mathbf{s}, \mathbf{h})$ over all eight latent histories $\mathbf{h} \in \mathcal{H}$:

$$L_i(\theta_0, \theta_1, \pi) = \sum_{\mathbf{h} \in \mathcal{H}} \Pr(\mathbf{s}_i \mid \mathbf{h}; \pi) \Pr(\mathbf{h}; \theta_0, \theta_1), \tag{1}$$

where $\Pr(\mathbf{s} \mid \mathbf{h}; \pi) = \prod_{t=1}^{3} \Pr(s_t \mid h_t; \pi)$ exploits the conditional independence of errors across waves, and $\Pr(\mathbf{h}; \theta_0, \theta_1)$ is the Markov chain probability of the latent history. The sample log-likelihood is $\ell = \sum_i \log L_i$.

The marginal likelihood (1) is a finite mixture: at each parameter draw, the data assign responsibility for each observation to the eight latent histories in proportion to their joint likelihood. This mixture structure is what makes the EM algorithm well-suited to this problem.

Identification

The three-wave observed employment distribution contains eight cell probabilities subject to one adding-up constraint, yielding seven free moments. The baseline model has three free parameters: θ_0, θ_1, and π (and, in the free initial-probability specification, a fourth parameter for $\Pr(h_1 = 1)$). Since seven moments exceed three (or four) free parameters, the model is over-identified when the true process satisfies the first-order Markov property.

The key identification insight is that the eight observed cell probabilities encode the Markov property as a testable restriction. Specifically, the conditional independence $s_1 \perp s_3 \mid s_2$ holds for any first-order Markov process on *observed* status—but it fails when the observed process is a noisy version of a latent Markov chain. In that case, knowing s_1 provides information about h_2 (because h_2 is correlated with h_1 through the latent dynamics), and hence information about s_3, even conditional on s_2. The magnitude of this additional dependence identifies π: if misclassification is zero, the observed process is itself Markov and satisfies the restriction; as π increases, the Markov violation grows. With three waves, the restriction is testable and π is non-parametrically identified up to the auxiliary assumption of a symmetric error structure.

EM algorithm

The EM algorithm maximises the complete-data log-likelihood—the log-likelihood of the full (s, h) data—by iterating between two steps. The E-step computes, for each individual and each latent history $\mathbf{h}$, the posterior responsibility:

$$\gamma_{i\mathbf{h}}^{(k)} = \frac{\Pr(\mathbf{s}_i \mid \mathbf{h}; \pi^{(k)}) \Pr(\mathbf{h}; \theta_0^{(k)}, \theta_1^{(k)})}{L_i(\theta_0^{(k)}, \theta_1^{(k)}, \pi^{(k)})}, \tag{2}$$

where k indexes the iteration. The M-step updates the parameters by maximising the expected complete-data log-likelihood:

$$\theta_0^{(k+1)} = \frac{\sum_i \sum_{\mathbf{h}:h_1=0,h_2=1} \gamma_{i\mathbf{h}}^{(k)} + \sum_i \sum_{\mathbf{h}:h_2=0,h_3=1} \gamma_{i\mathbf{h}}^{(k)}}{\sum_i \sum_{\mathbf{h}:h_1=0} \gamma_{i\mathbf{h}}^{(k)} + \sum_i \sum_{\mathbf{h}:h_2=0} \gamma_{i\mathbf{h}}^{(k)}}, \tag{3}$$

$$\theta_1^{(k+1)} = \frac{\sum_i \sum_{\mathbf{h}:h_1=1,h_2=0} \gamma_{i\mathbf{h}}^{(k)} + \sum_i \sum_{\mathbf{h}:h_2=1,h_3=0} \gamma_{i\mathbf{h}}^{(k)}}{\sum_i \sum_{\mathbf{h}:h_1=1} \gamma_{i\mathbf{h}}^{(k)} + \sum_i \sum_{\mathbf{h}:h_2=1} \gamma_{i\mathbf{h}}^{(k)}}, \tag{4}$$

$$\pi^{(k+1)} = \frac{\sum_i \sum_{\mathbf{h}} \gamma_{i\mathbf{h}}^{(k)} \sum_{t=1}^{3} \mathbf{1}[s_{it} \neq h_t]}{3 \sum_i 1}. \tag{5}$$

The M-step updates (3)–(22) have a transparent interpretation: θ_0 is the responsibility-weighted empirical fraction of periods in which a truly non-employed worker becomes employed, θ_1 is the analogous exit fraction, and π is the responsibility-weighted empirical misclassification rate. Each update is in closed form, so the algorithm requires no numerical optimisation and is guaranteed to converge to a local maximum of the observed-data log-likelihood (Dempster et al., 1977).

Standard errors are obtained by a parametric bootstrap with $B = 200$ repetitions, sampling from the multinomial distribution over the eight observed histories at the estimated cell probabilities.

3.2 Extensions

The extensions below each generalise one aspect of the baseline model. All retain the core EM structure; the main difference is in the M-step update for the extended parameter set.

3.2.1 Observable worker heterogeneity

We allow the entry and exit probabilities to depend on individual characteristics X_i (age, education, gender, race, contract type) through probit link functions:

$$\theta_{0i} = \Phi(X_i\beta_0), \qquad \theta_{1i} = \Phi(X_i\beta_1),$$

where $\Phi(\cdot)$ is the standard normal CDF. The misclassification parameter π is held common across individuals in the main specification; a variant allows π to vary with observable reliability indicators (see Section 3.2.3). Since the M-step for β_0 and β_1 no longer has a closed form, we use the Generalised EM (GEM) algorithm of Dempster et al. (1977), performing a single gradient step in the direction of the expected complete-data likelihood gradient at each iteration. The GEM algorithm is guaranteed to produce non-decreasing likelihoods and converges under the same conditions as the full EM.

3.2.2 Finite mixture model for unobserved heterogeneity

Unobserved worker heterogeneity—workers who persistently have very high or very low transition probabilities—can mimic the Markov violations created by misclassification: a persistently high-exit worker looks, in the observed data, like someone who repeatedly enters and exits employment within the observation window. We disentangle these mechanisms through a two-type finite mixture model (FMM):

$$L_i(\phi, \theta^A, \theta^B, \pi) = \phi\, L_i(\theta^A, \pi) + (1 - \phi)\, L_i(\theta^B, \pi),$$

where type A has parameters (θ_0^A, θ_1^A), type B has parameters (θ_0^B, θ_1^B), and ϕ is the mixing proportion. The EM algorithm for the FMM introduces a second layer of latent variables (the type indicator) and iterates the E-step responsibilities jointly over latent states and types. The misclassification parameter π is identified separately from the type heterogeneity because Markov violations attributable to types can be attributed to heterogeneity in $(\theta^A, \theta^B, \phi)$, while additional violations residual to the type structure are attributed to π.

3.2.3 Inconsistency-augmented misclassification

If misclassification is driven by survey response error, respondents who make other types of survey errors should also have higher misclassification rates. We test this prediction by allowing π to depend on an indicator of survey unreliability. Specifically, we define:

- $A_i = 1$ if individual i has an age inconsistency in any wave pair;
- $E_i = 1$ if individual i has an education inconsistency in any wave pair.

We then estimate the generalised model:

$$\pi_i = \text{logit}^{-1}(\beta_0 + \beta_1 A_i + \beta_2 E_i),$$

where $\text{logit}^{-1}(x) = e^x/(1 + e^x)$ ensures that $\pi_i \in (0, 1)$. The identifying variation comes from whether the inconsistency is attributed to the first, second, or third wave of the triad. We use a wave-attribution rule based on the sign of the change in the inconsistent variable to assign each inconsistency to the wave most likely responsible, and we then condition π_i on the presence of a second-wave attribution (since a first-wave error in the first-wave status and a third-wave error in the third-wave status would be absorbed into the initial and terminal probabilities). The details of the attribution rule are described in Appendix A.

3.2.4 AR(2) extension

The baseline model assumes that the latent true employment process is first-order Markov: conditional on last quarter's true status, the probability of this quarter's status is independent of all earlier history. This rules out duration dependence—the possibility that the probability of finding a job increases or decreases with the length of the current non-employment spell. We relax this assumption by extending the model to a second-order Markov specification using four consecutive waves.

In the AR(2) model, the latent transition probability depends on the current state and the state one period earlier:

$$p_{00} = \Pr(h_t = 1 \mid h_{t-1} = 0, h_{t-2} = 0), \quad p_{01} = \Pr(h_t = 1 \mid h_{t-1} = 0, h_{t-2} = 1),$$
$$p_{10} = \Pr(h_t = 0 \mid h_{t-1} = 1, h_{t-2} = 0), \quad p_{11} = \Pr(h_t = 0 \mid h_{t-1} = 1, h_{t-2} = 1),$$

where the first subscript denotes the state at $t-1$ and the second the state at $t-2$. This specification accommodates negative duration dependence in non-employment (if $p_{00} > p_{01}$, a longer non-employment spell reduces the probability of entry) and positive duration dependence in employment. The four-wave observed histories, of which there are 16, support identification of these four additional transition probabilities along with the misclassification parameter. The full derivation is in Appendix B.

3.2.5 Tenure contamination model

The baseline model and its extensions above identify misclassification from Markov violations in the employment status sequence. The tenure contamination model provides an independent identification channel using job tenure and non-employment duration data.

The key observation is that if an employment observation is misclassified—say, a truly non-employed worker is recorded as employed at wave t—the reported job tenure at wave $t+1$ should be inconsistent with normal tenure accumulation.

Specifically, if the worker was truly non-employed at t and is truly employed at $t+1$, their tenure at $t+1$ should be short (consistent with a new job). But if the worker is also truly employed at $t-1$, their tenure at $t-1$ should be long (consistent with a continuing job), and the jump in tenure between $t-1$ and $t+1$ (or the failure of tenure to accumulate between t and $t+1$) signals a "contaminated" intermediate observation.

We formalise this intuition as follows. Let d_{it} denote the duration variable (tenure if employed, time since last job if non-employed) at wave t. We model the joint distribution of $(s_{i,t-1}, d_{i,t-1}, s_{it}, d_{it}, s_{i,t+1}, d_{i,t+1})$ as a function of the underlying latent employment sequence $\mathbf{h}$ and an additional contamination parameter ε: the probability that a duration observation is "contaminated," meaning it is drawn from the distribution of the wrong employment state. The contamination probability is interpreted as a form of tenure recall error that is correlated with employment state misclassification. Jointly identifying π and ε from the observed employment–tenure sequence requires an assumption about the marginal distribution of tenure; we model tenure as exponential with rate parameters (λ_g, λ_d) for employed and non-employed workers respectively, and fit the model by EM. The full derivation is in Appendix C.

3.3 Matching and sample construction

Each three-wave triad is constructed by linking panel observations using household and individual identifiers. We consider a hierarchy of matching strictness, from the baseline matching—which requires agreement on household identifiers only—to stricter matching algorithms that additionally require agreement on gender, race, and certain time-invariant characteristics. Stricter matching reduces the sample size (by discarding potential false matches) but should also reduce the incidence of spurious transitions arising from mismatched observations rather than genuine employment changes. As discussed in Section 5, we use this variation as a design-based robustness check: if our estimated misclassification π reflects genuine survey error rather than modelling artefact, stricter matching should reduce but not eliminate π.

4 Main Results

4.1 Baseline estimates

Table 3 reports implied entry and exit probabilities from the baseline model, and Table 4 reports the underlying structural parameter estimates. The columns cover the six specification variants: stationarity-restricted and free initial probability, crossed with no misclassification, symmetric misclassification, and asymmetric misclassification.

Without misclassification correction—the standard Markov transition matrix applied to observed employment status—the estimated quarterly entry rate is 8.28% and the exit rate is 8.54%. These estimates are close to the raw transition rates in Table 1, confirming that the no-misclassification specification simply recovers the moments of the observed transition matrix.

The symmetric misclassification specification yields a dramatically different picture. The estimated misclassification probability is $\hat{\pi} = 2.98\%$, and the implied entry and exit rates fall to 2.93% and 3.00% respectively. The correction factor is approximately three: a single additional parameter reduces the estimated transition rates by two-thirds. This is the main finding of the paper. The factor-of-three correction is driven by the severity of the Markov violation documented in Section 2.2: even a small misclassification probability inflates high-frequency transition rates substantially because each transient error fabricates two transitions.

The asymmetric misclassification specification, which allows the probabilities of falsely reporting employment and non-employment to differ, produces estimates that are virtually indistinguishable from the symmetric specification. The estimated misclassification probabilities for both directions are approximately 3%, and the implied transition rates are within a rounding error of the symmetric case. The symmetric restriction is not rejected and we adopt it in all subsequent specifications for parsimony.

Comparing the stationarity-restricted and free initial-probability specifications yields similarly negligible differences. The data are consistent with the estimated Markov chain having been in the stationary distribution at the start of the observation window, and the stationarity assumption is maintained in subsequent models.

The standard errors, obtained from 200 parametric bootstrap replications, indicate that the misclassification probability is estimated with considerable precision. The 95% bootstrap confidence interval for $\hat{\pi}$ in the symmetric stationarity-restricted specification is approximately [2.5%, 3.4%], comfortably above zero and well below magnitudes that would be necessary to explain the raw transition matrix moments without correction.

4.2 Observable worker heterogeneity

A natural concern is that the estimated misclassification probability is capturing heterogeneity in workers' true transition rates rather than measurement error. Workers who are persistently close to the employment margin may exhibit frequent transitions in the data even if the Markov assumption holds; if these workers are a small share of the sample, their high transition rates could generate Markov violations when the average transition rate is treated as homogeneous.

Table 5 reports the implied transition rates from the observable heterogeneity extension. We consider three covariate sets of increasing richness: (1) age and education only; (2) additionally controlling for gender, race, and urban/rural status; (3) additionally controlling for contract type and sector. Allowing transition probabilities to vary with these characteristics substantially affects the composition of the entry and exit hazards—the average implied entry and exit rates shift as the model accounts for the composition of the sample across covariate cells. Nonetheless, the estimated misclassification probability remains economically and statistically significant in all three specifications, at 2.36%, 2.23%, and 1.17% respectively. Even the most flexible covariate specification, which allows transition rates to differ by gender, race, age, education, urban status, contract type, and sector, cannot account for the Markov violations in the data without retaining a meaningful misclassification probability. The conclusion that conventional transition matrices overstate true transition rates by a factor of approximately two to three is robust to controlling for observable worker heterogeneity.

4.3 Unobserved worker heterogeneity

The finite mixture model (FMM) extension relaxes the assumption that workers form a homogeneous population with respect to their underlying transition probabilities. Table 6 reports the FMM results.

Without accounting for misclassification, the two-type FMM assigns approximately equal weight to each type ($\hat{\phi} \approx 0.50$). Type B has very high estimated entry and exit rates ($\hat{\theta}_0^B \approx 0.37$, $\hat{\theta}_1^B \approx 0.57$), while type A has much lower rates. This "coin-flip" type B is an implausible description of any group of actual workers; it is more naturally interpreted as a misclassification artefact, where a subset of observations with particularly noisy employment signals is assigned to a spurious type.

Adding misclassification to the FMM substantially alters this picture. The mixing proportion shifts to $\hat{\phi} = 43.26\%$, the type A entry and exit rates fall to 6.05% and 1.95%, and the type B rates become 2.13% and 5.44%. These are plausible transition rates for two distinct worker groups—perhaps corresponding to workers with stable versus unstable employment relationships. The estimated misclassification probability in the FMM with π is $\hat{\pi} = 2.92\%$, virtually identical to the baseline. This is the key result: allowing the model to explain Markov violations through either unobserved heterogeneity or misclassification, it attributes approximately the same fraction to misclassification as the simpler homogeneous model. Unobserved heterogeneity does not absorb the misclassification signal; instead, adding misclassification resolves the implausible coin-flip type.

4.4 Inconsistency-augmented misclassification

Table 7 reports the results of the inconsistency-augmented model. If misclassification in employment status is driven by the same underlying propensity for survey response error that generates age and education inconsistencies, then conditioning misclassification on inconsistency indicators should significantly improve the model fit, and the coefficient on the inconsistency indicators should be positive and significant.

The results confirm this prediction strongly. The baseline misclassification probability—for respondents with no age or education inconsistency—is estimated at $\hat{\pi}_0 = 2.22\%$, slightly lower than the homogeneous baseline estimate. An age inconsistency in any wave pair raises the estimated misclassification probability by 15.72 percentage points, and an education inconsistency raises it by 4.22 percentage points, both statistically significant at conventional levels. The implied mean misclassification probability across the full sample, integrating over the distribution of inconsistency indicators, is 3.40%.

These results support the interpretation that misclassification is driven by genuine survey measurement error rather than by model misspecification. Respondents who produce objectively incorrect responses to verifiable questions (age and education) are substantially more likely to produce misclassification-consistent employment dynamics. This mechanism evidence is independent of the employment status data itself and cannot be explained by heterogeneity in true transition rates.

5 Alternative Specifications

5.1 AR(2): second-order Markov dynamics

The baseline model assumes that the latent employment process is first-order Markov. If employment exhibits genuine duration dependence—if the probability of finding a job declines with the length of a non-employment spell, or if the probability of losing a job declines with seniority—then the first-order Markov restriction could be misspecified, and the EM estimator might attribute duration-dependent dynamics to misclassification.

Tables 8 and 9 report the results of the AR(2) extension, which uses four

Table 3: Baseline EM model: implied probabilities (%)

	(1)	(2)	(3)	(4)	(5)
	No miscl.		Symmetric		A
	Stat.	Free	Stat.	Free	Stat.
Entry rate (θ_0)	8.28**	8.28**	2.93**	2.99**	2.93**
(0.05)	(0.04)	(0.04)	(0.06)	(0.04)	(0.05)
Exit rate ($1-\theta_1$)	8.54**	8.54**	3.00**	2.92**	3.00**
(0.05)	(0.05)	(0.04)	(0.06)	(0.04)	(0.05)
Employment rate (α^*)	49.23**	49.23**	49.47**	50.57**	49.48**
(0.21)	(0.18)	(0.27)	(0.53)	(0.26)	(0.52)
π (misclassification)	—	—	2.98**	2.99**	—
(—)	(—)	(0.03)	(0.03)	(—)	(—)
π_0 (false positive)	—	—	—	—	2.98**
(—)	(—)	(—)	(—)	(0.04)	(0.04)
π_1 (false negative)	—	—	—	—	2.99**
(—)	(—)	(—)	(—)	(0.04)	(0.04)
N	402,923	402,923	402,923	402,923	402,923

Note: Estimates in %. Bootstrap SE in parentheses ($B = 200$). Employment rate is steady-state $\alpha^* = \theta_0/(\theta_0 + 1 -$

consecutive waves and allows the transition probabilities to depend on both the current and lagged true employment state. The four-wave linked sample contains approximately 317,000 observations, somewhat smaller than the three-wave sample because a fourth valid wave is required.

The AR(2) estimates reveal substantial and statistically significant duration dependence. Among non-employed workers, those who were also non-employed in the prior quarter have a true entry probability of only 4.33%, compared with 30.61% for those who were employed in the prior quarter. This large difference ($\hat{\theta}_{01} - \hat{\theta}_{00} = 26.28$ percentage points) indicates strong negative duration dependence in non-employment: workers who have been non-employed for at least two consecutive quarters are much less likely to find a job in the current quarter than those who recently entered non-employment. Among employed workers, those who were also employed in the prior quarter have a separation probability of only 4.19%, compared with 24.57% for those who were non-employed in the prior quarter, indicating that recent entrants into employment are much more likely to

Table 4: Baseline EM model: parameter estimates

	(1)	(2)	(3)	(4)	(5)	(6)
	No miscl.		Symmetric		Asymmetric	
	Stat.	Free	Stat.	Free	Stat.	Free
Transition parameters						
θ_0 (entry)	0.0828**	0.0828**	0.0293**	0.0299**	0.0293**	0.0299**
(0.0005)	(0.0004)	(0.0004)	(0.0006)	(0.0004)	(0.0005)	
θ_1 (persistence)	0.9146**	0.9146**	0.9700**	0.9708**	0.9700**	0.9708**
(0.0005)	(0.0005)	(0.0004)	(0.0006)	(0.0004)	(0.0005)	
α (initial empl.)	0.4923**	0.4852**	0.4947**	0.4837**	0.4948**	0.4832**
(0.0021)	(0.0009)	(0.0027)	(0.0009)	(0.0026)	(0.0011)	
Misclassification						
π (symmetric)	—	—	0.0298**	0.0299**	—	—
(—)	(—)	(0.0003)	(0.0003)	(—)	(—)	
π_0 (false positive)	—	—	—	—	0.0298**	0.0305**
(—)	(—)	(—)	(—)	(0.0004)	(0.0004)	
π_1 (false negative)	—	—	—	—	0.0299**	0.0292**
(—)	(—)	(—)	(—)	(0.0004)	(0.0004)	
Log-likelihood	-357.2M	-357.2M	-349.4M	-349.4M	-349.4M	-349.4M
N	402,923	402,923	402,923	402,923	402,923	402,923

Note: Bootstrap SE in parentheses ($B = 200$). Significance: ** $p < 0.01$, * $p < 0.05$, · $p < 0.10$.

Table 5: Covariate EM models: implied probabilities (stationary α)

	(1)	(2)	(3)	(4)	(5)	(6)
		No miscl.			Symmetric	
	Set 1	Set 2	Set 3	Set 1	Set 2	Set 3
Mean entry rate (%)	9.05**	9.34**	16.18**	4.21**	4.59**	9.94**
(0.05)	(0.05)	(0.18)	(0.04)	(0.04)	(0.30)	
Mean exit rate (%)	10.98**	11.47**	14.36**	5.14**	5.71**	10.11**
(0.07)	(0.07)	(0.10)	(0.06)	(0.06)	(0.17)	
Mean employment rate (%)	47.69**	47.32**	49.19**	47.63**	47.31**	48.79**
(0.16)	(0.15)	(0.16)	(0.24)	(0.20)	(0.20)	
π misclassification (%)	—	—	—	2.36**	2.23**	1.17**
(—)	(—)	(—)	(0.02)	(0.02)	(0.04)	
Log-likelihood	-318.4M	-312.6M	-257.9M	-313.6M	-308.1M	-264.9M
N	394,823	394,823	394,823	394,823	394,823	394,823
Covariates included						
Age, Age^2, Educ.	Yes	Yes	Yes	Yes	Yes	Yes
+ Race, Gender	No	Yes	Yes	No	Yes	Yes
+ Contract type	No	No	Yes	No	No	Yes

Note: Estimates in %. Bootstrap SE in parentheses ($B = 200$). Significance: ** $p < 0.01$, * $p < 0.05$, $^{\cdot}$ $p < 0.10$.

Table 6: Finite mixture model: implied probabilities

	(1)	(2)	(3)	(4)
	No miscl.			Symmetric
	Stat.	Free	Stat.	Free
Type A (stable workers)				
Entry rate θ_0^A (%)	99.90	55.96**	6.05**	5.81**
(0.00)	(0.64)	(1.23)	(0.15)	
Exit rate $1-\theta_1^A$ (%)	2.72**	5.25**	1.95**	0.19**
(0.07)	(0.05)	(0.67)	(0.04)	
Employment rate α^A (%)	97.35**	91.43**	75.61**	96.88**
(0.07)	(0.13)	(8.47)	(0.61)	
Type B (unstable workers)				
Entry rate θ_0^B (%)	5.98**	3.28**	2.13**	0.33**
(0.09)	(0.04)	(0.36)	(0.09)	
Exit rate $1-\theta_1^B$ (%)	47.32**	90.11**	5.44**	17.37**
(2.24)	(0.61)	(1.17)	(2.42)	
Employment rate α^B (%)	11.21**	3.52**	28.17**	1.88**
(0.60)	(0.05)	(10.04)	(0.54)	
Mixture and misclassification				
ϕ prob. type A (%)	42.83**	50.86**	43.26*	64.45**
(0.37)	(0.10)	(20.52)	(0.60)	
Wtd. avg. entry rate (%)	46.21**	30.07**	3.83**	3.87**
(0.31)	(0.31)	(0.47)	(0.09)	
Wtd. avg. exit rate (%)	28.21**	46.95**	3.93**	6.30**
(1.15)	(0.31)	(0.46)	(0.69)	
π misclassification (%)	—	—	2.92**	3.01**
(—)	(—)	(0.03)	(0.03)	
Log-likelihood	-341.0M	-340.9M	-341.0M	-340.9M
N	394,823	394,823	394,823	394,823

Note: Estimates in %. Type A has higher θ_1 (more stable). Bootstrap SE in parentheses ($B = 200$). Significance: *

Table 7: Inconsistency-augmented EM model: implied probabilities

	(1) Stationary	(2) Free α
Transition parameters		
Entry rate θ_0 (%)	2.50^{**}	2.60^{**}
(0.04)	(0.05)	
Exit rate $1-\theta_1$ (%)	2.43^{**}	2.21^{**}
(0.05)	(0.05)	
Employment rate α^* (%)	50.72^{**}	54.15^{**}
(0.29)	(0.65)	
Misclassification (logistic link)		
π^{base} (%)	2.22^{**}	2.25^{**}
(0.03)	(0.03)	
Add'l miscl. (age incons.) (pp)	15.72^{**}	15.73^{**}
(0.25)	(0.27)	
Add'l miscl. (educ. incons.) (pp)	4.22^{**}	4.23^{**}
(0.11)	(0.11)	
Mean $\bar{\pi}$ (Pop. avg. rate) (%)	3.40^{**}	3.44^{**}
(0.03)	(0.03)	
Log-likelihood	-334.1M	-333.6M
N	394,823	394,823

Note: Implied rates in %, additional effects in pp. Bootstrap SE ($B = 200$). Significance: ** $p < 0.01$, * $p < 0.05$, $^{\cdot}$

exit in the subsequent quarter.

Despite this rich duration dependence, the estimated misclassification probability in the AR(2) model is $\hat{\pi} = 1.14\%$—positive, statistically significant, and large enough to imply meaningful correction to the observed transition matrix. Although this estimate is lower than the baseline 2.98%, the persistence of a positive misclassification probability confirms that duration dependence alone cannot explain the Markov violations in the data. The two mechanisms—duration dependence and misclassification—are jointly identified and coexist in the AR(2) specification.

The AR(2) results also imply that the strong negative duration dependence in the baseline model may be partly an artefact of the first-order Markov restriction: once longer-order dynamics are accommodated, the apparent duration-dependence-from-misclassification confound is resolved, and both mechanisms have distinct empirical signatures.

5.2 Tenure contamination model

Tables 10, 11, and 12 report the results of the tenure contamination model. This specification exploits the joint distribution of employment status and job tenure to provide an independent structural identification channel for misclassification.

The estimated contamination probability is $\hat{\varepsilon} = 24.23\%$: nearly one in four tenure observations in the sample is estimated to be "contaminated," in the sense that it is drawn from the wrong employment state's tenure distribution. This high contamination rate reflects both genuine tenure recall error and the structural implications of employment state misclassification for flanking tenure observations. Given this contamination rate, the estimated misclassification probability is $\hat{\pi} = 8.64\%$ under the static specification and somewhat higher under the CTMC-linked specification.

The tenure contamination model recovers mean employment spell lengths from the estimated transition rates. Table 12 reports that the implied mean employment spell length is approximately 5.85 years (70.2 months), with $P(\text{clock-consistent}) =$

75.77%—the share of employment-tenure sequences in the sample that are consistent with normal tenure clock accumulation (i.e., where tenure increases by approximately one quarter per wave). This proportion substantially exceeds what would be expected if every employment observation were drawn independently; the high clock-consistency rate validates the exponential duration model as a reasonable approximation.

We note that the misclassification estimate from the tenure contamination model is higher than the baseline. This is partly a consequence of the longer observation window (tenure flanks span multiple waves) and partly because the exponential duration assumption introduces additional functional form restrictions. We interpret the tenure contamination results as corroborating the direction and approximate magnitude of the baseline finding—misclassification is present, economically substantial, and structurally identified—rather than as a precise alternative estimate of π.

5.3 Matching algorithm sensitivity

Table 13 reports sensitivity of the results to the strictness of the wave-to-wave matching algorithm. The baseline matching algorithm links individuals using household and person identifiers only. The "stricter" matching algorithm additionally requires agreement on gender, race, and a time-invariant household characteristic; the "strictest" algorithm imposes further agreement on an additional set of stable characteristics. Stricter matching reduces the sample size by discarding potential false matches (i.e., observations that look like the same individual across waves but may be different individuals who were incorrectly linked by the identifier algorithm).

As matching becomes stricter, both the raw (no misclassification) transition rates and the EM-corrected transition rates decline, and the estimated misclassification probability falls from $\hat{\pi} = 2.82\%$ under the baseline matching to $\hat{\pi} = 2.25\%$ under the strictest matching. This pattern is consistent with the interpretation that some of the Markov violations attributed to misclassification under the baseline matching reflect genuine mismatches: when a non-employed person is incorrectly linked to an employed observation from another household member, the resulting spurious transition inflates measured entry and exit rates and creates apparent Markov violations that the EM algorithm would otherwise attribute to misclassification.

Crucially, however, even the strictest matching algorithm retains a substantial misclassification probability of 2.25%. Matching errors are neither the primary nor the sole source of the Markov violations in the data; genuine survey measurement error in employment status contributes independently. This design-based robustness check, which varies identification assumptions (how the panel is constructed) rather than the structural model, strongly supports the main finding.

The implied entry and exit rates under the strictest matching and symmetric misclassification correction (2.36% and 2.25% respectively) are slightly lower than the baseline 2.93% and 3.00%, but the overall conclusion is unchanged: true employment transition rates in South Africa are approximately three times lower than the raw transition matrix would suggest, and employment is considerably more persistent than previously characterised.

6 Conclusion

Classification error in survey responses to employment questions is pervasive and, even when small, systematically inflates measured employment transition rates. We have developed a structural EM-algorithm estimator that jointly identifies true transition rates and misclassification probabilities from three consecutive waves of panel data, requiring no reinterview data or administrative validation. The estimator exploits the fact that transient misclassification causes the observed employment sequence to violate the first-order Markov property in a specific, quantifiable way, and converts this insight into a fast, numerically reliable sequence of closed-form parameter updates.

Applied to the South African Quarterly Labour Force Survey, the estimator yields a misclassification probability of approximately 3%, which reduces estimated quarterly entry and exit rates from roughly 8% to roughly 3%—a cor-

Table 8: AR(2) EM model: implied transition probabilities (%)
p_{jk} = Pr(employed in period t | status in $t-2$ was j, status in $t-1$ was k)

	(1) No miscl.	(2) Symmetric π	(3) Asymmetric (π_0, π_1)
Transition probabilities			
p_{00} (%)	5.49**	4.33**	4.33**
	(0.17)	(0.22)	(0.22)
p_{10} (%)	35.36**	30.61**	30.58**
	(1.40)	(1.93)	(1.78)
p_{01} (%)	65.82**	71.24**	71.21**
	(1.17)	(2.30)	(1.89)
p_{11} (%)	94.74**	95.81**	95.81**
	(0.17)	(0.20)	(0.18)
Steady state			
Employment rate α^* (%)	51.42**	51.42**	51.43**
	(1.30)	(1.56)	(1.80)
Misclassification			
π (%)	—	1.14**	—
	(—)	(0.15)	(—)
π_0 (%)	—	—	1.14**
	(—)	(—)	(0.18)
π_1 (%)	—	—	1.14**
	(—)	(—)	(0.15)
Log-likelihood	-0.0M	-0.0M	-0.0M
N	19,093	19,093	19,093

Note: Bootstrap standard errors (B = 200) in parentheses. Stars: $^\cdot$ p<0.10, * p<0.05, ** p<0.01 (two-sided z-test, I

Table 9: AR(2) EM model: parameter estimates

	(1) No miscl.	(2) Symmetric π	(3) Asymmetric (π_0, π_1)
θ_0	0.0549^{**}	0.0433^{**}	0.0433^{**}
	(0.0017)	(0.0022)	(0.0022)
θ_{01}	0.2987^{**}	0.2627^{**}	0.2624^{**}
	(0.0142)	(0.0190)	(0.0174)
θ_1	0.0526^{**}	0.0419^{**}	0.0419^{**}
	(0.0017)	(0.0020)	(0.0018)
θ_{10}	0.2892^{**}	0.2457^{**}	0.2460^{**}
	(0.0119)	(0.0227)	(0.0184)
Misclassification			
π	0.0000	0.0114^{**}	—
	(—)	(0.0015)	(—)
π_0	—	—	0.0114^{**}
	(—)	(—)	(0.0018)
π_1	—	—	0.0114^{**}
	(—)	(—)	(0.0015)
Log-likelihood	-0.0M	-0.0M	-0.0M
N	19,093	19,093	19,093

Note: Bootstrap standard errors (B = 200) in parentheses. Stars: $\cdot$ p<0.10, * p<0.05, ** p<0.01 (two-sided z-test, I

Table 10: EM tenure contamination model: implied probabilities (%).
Employment rate is steady-state $\alpha^* = \theta_0/(\theta_0 + 1 - \theta_1)$.

	(1) Free (7 param.)	(2) Stationary (6 param.)	(3) CTMC-linked (5 param.)	(4) Stat.+linked (4 param.)
Labour-market flow rates				
Entry rate (θ_0, %)	9.92**	9.85**	4.08**	4.14**
(0.06)	(0.06)	(0.01)	(0.00)	
Exit rate ($1 - \theta_1$, %)	8.17**	8.31**	4.26**	4.26**
(0.05)	(0.04)	(0.01)	(0.01)	
Employment rate (α^*, %)	54.83**	54.26**	48.93**	49.28**
(0.21)	(0.10)	(0.10)	(0.03)	
Measurement error rates				
π: misclassification (%)	8.68**	8.64**	10.21**	10.02**
(0.04)	(0.04)	(0.05)	(0.05)	
ε: contamination (%)	24.20**	24.23**	24.79**	24.69**
(0.09)	(0.07)	(0.08)	(0.07)	
N	402,923	402,923	402,923	402,923

Note: All quantities in %. Bootstrap SE in parentheses ($B = 200$). Significance: ** $p < 0.01$, * $p < 0.05$, $^{\cdot}$ $p < 0.10$.

Table 11: EM tenure contamination model: parameter estimates. Preferred specification: column (4) (stationary + CTMC-linked).

	(1) Free (7 param.)	(2) Stationary (6 param.)	(3) CTMC-linked (5 param.)	(4) Stat.+linked (4 param.)
Transition parameters				
θ_0 (entry)	0.0992**	0.0985**	0.0408**	0.0414**
(0.0006)	(0.0006)	(0.0001)	(0.0000)	
θ_1 (persistence)	0.9183**	0.9169**	0.9574**	0.9574**
(0.0005)	(0.0004)	(0.0001)	(0.0001)	
α (initial empl.)	0.5322**	—	0.5498**	—
(0.0009)		(0.0010)		
Misclassification & contamination				
π (misclassification)	0.0868**	0.0864**	0.1021**	0.1002**
(0.0004)	(0.0004)	(0.0005)	(0.0005)	
ε (contamination)	0.2420**	0.2423**	0.2479**	0.2469**
(0.0009)	(0.0007)	(0.0008)	(0.0007)	
Exponential spell rates				
λ_g (employment rate)	0.1710**	0.1710**	—	—
(0.0005)	(0.0004)			
λ_d (non-employment rate)	0.1584**	0.1584**	—	—
(0.0005)	(0.0004)			
Log-likelihood	-1747.4M	-1747.5M	-1752.7M	-1754.2M
N	402,923	402,923	402,923	402,923

Note: Bootstrap SE in parentheses ($B = 200$). Significance: ** $p < 0.01$, * $p < 0.05$, $^{\cdot}$ $p < 0.10$.

Table 12: EM tenure contamination model: tenure and spell-length interpretation. Durations derived from $\hat{\lambda}_g$ and $\hat{\lambda}_d$ assuming Exponential distributions. Contamination probabilities derived from $\hat{\varepsilon}$.

	(1) Free (7 param.)	(2) Stationary (6 param.)	(3) CTMC-linked (5 param.)
Employment spell duration ($\hat{\lambda}_g$)			
Mean spell (years)	5.85^{**}	5.85^{**}	5.74^{**}
(0.02)	(0.01)	(0.02)	(0.01)
Mean spell (months)	70.2^{**}	70.2^{**}	68.9^{**}
(0.2)	(0.2)	(0.2)	(0.1)
Median spell (months)	48.6^{**}	48.6^{**}	47.8^{**}
(0.1)	(0.1)	(0.1)	(0.1)
Non-employment spell duration ($\hat{\lambda}_d$)			
Mean spell (years)	6.31^{**}	6.31^{**}	6.00^{**}
(0.02)	(0.02)	(0.02)	(0.01)
Mean spell (months)	75.8^{**}	75.8^{**}	72.0^{**}
(0.2)	(0.2)	(0.2)	(0.1)
Tenure measurement quality ($\hat{\varepsilon}$)			
P(clock-consistent), $1-\varepsilon$ (%)	75.80^{**}	75.77^{**}	75.21^{**}
(0.09)	(0.07)	(0.08)	(0.07)
P(spell-pair both clean), $(1-\varepsilon)^2$ (%)	57.45^{**}	57.41^{**}	56.57^{**}
(0.14)	(0.10)	(0.12)	(0.10)
P(K=3 all clean), $(1-\varepsilon)^3$ (%)	43.54^{**}	43.50^{**}	42.55^{**}
(0.16)	(0.12)	(0.14)	(0.11)
Contaminated per 1,000 observations	242.0^{**}	242.3^{**}	247.9^{**}
(0.9)	(0.7)	(0.8)	(0.7)
N	402,923	402,923	402,923

Note: Spell durations are $1/\hat{\lambda}$ (mean) and $\log(2)/\hat{\lambda}$ (median) under the Exp($\hat{\lambda}$) model. Bootstrap SE in parentheses

Table 13

	(1)	(2)	(3)	(4)
Entry rate	7.84***	6.50***	2.61***	2.36***
	(0.13)	(0.12)	(0.18)	(0.16)
Exit rate	7.74***	6.68***	2.58***	2.44***
	(0.13)	(0.12)	(0.18)	(0.16)
Misclassification rate			2.82***	2.25***
			(0.12)	(0.11)
Misclassification	No	No	Yes	Yes
Matching algorithm	Stricter	Strictest	Stricter	Strictest
LL	-28480.3	-27064.8	-27823.1	-26490
Observations	21,932	17,737	21,932	17,737

Note: *p<0.1; **p<0.05; ***p<0.01

rection factor of nearly three. This finding is robust across all specifications we consider: controlling for observable worker characteristics (age, education, gender, race, contract type) reduces the estimated misclassification probability moderately but does not eliminate it; allowing unobserved two-type heterogeneity in transition rates leaves the misclassification estimate essentially unchanged while resolving an implausible “coin-flip” high-churn type; extending the Markov order to AR(2) with four waves reveals substantial duration dependence but retains a positive misclassification probability; and the tenure contamination model provides an independent structural corroboration using job tenure data. Survey respondents with objectively inconsistent age or education progressions exhibit employment dynamics consistent with misclassification rates more than fifteen percentage points higher than reliable respondents, providing direct mechanism evidence that the estimated misclassification reflects genuine survey error.

The policy implications are substantial. If true quarterly entry and exit rates are approximately 3%, the average employment spell in South Africa is on the order of several years—considerably longer than the raw transition matrix would suggest. This characterisation of South African employment as highly persistent implies that unemployment reflects primarily a failure of job creation rather than a failure of job retention. The policy prescription is correspondingly different: interventions that lower barriers to hiring, promote labour demand, and reduce the cost of formal employment are more likely to be effective than policies focused on worker protections, employment insurance, or the facilitation of job search. This does not imply that active labour market policies are irrelevant—workers who are persistently non-employed may still benefit from training and placement programmes—but it does imply that the design of such programmes should be predicated on a realistic understanding of the degree of labour market dynamism.

Beyond South Africa, our estimator is applicable wherever three-wave panel data on a binary state are available and reinterview data are not. The measurement error problem we address is particularly acute in developing country surveys, where reinterviews are uncommon and measurement quality may be lower, but it arises in any survey context where respondents may conflate employment states or where the survey instrument introduces transient coding errors. Potential extensions include applications to poverty transitions (Bigsten and Shimeles, 2008), health insurance coverage (van Os and Guloksuz, 2017), and educational enrolment (Fredriksson, 1997), all of which involve binary state variables measured with potential transient error in panel surveys. The misclassification problem is not unique to employment, and the EM approach extends naturally to any setting where the first-order Markov assumption can be motivated for the underlying true process.

Several limitations of our approach merit acknowledgment. First, the baseline misclassification structure assumes symmetric and time-invariant error probabilities. The inconsistency-augmented specification relaxes the homogeneity assumption, but the asymmetric baseline results suggest that the symmetry restriction is innocuous in our data. Second, we have treated the first-order Markov assumption on the latent true process as the baseline identification assumption, relaxing it only in the AR(2) extension. Higher-order dynamics, time-varying transition rates, or more complex dependence structures could in principle be accommodated within the EM framework at the cost of additional data requirements. Third, our approach identifies misclassification from the time-series variation in employment status across waves; it cannot separately identify misclassification that is perfectly correlated across waves (a worker who is always misclassified would appear to have a different true state, not a noisy version of their actual state). Future work could extend the approach to longer panels, time-varying misclassification probabilities, or multivariate employment state variables (employed, unemployed, inactive), all of which are within the scope of the EM framework developed here.

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A Baseline EM algorithm: full derivation

A.1 Model

A.1.1 States and notation

Consider a population observed at three equally spaced waves $t \in \{1, 2, 3\}$, separated by quarterly intervals of $\Delta = 0.25$ years. Let $h_t \in \{0, 1\}$ denote the *true* (latent) employment state at wave t, where 1 denotes employment and 0 denotes nonemployment, and let $s_t \in \{0, 1\}$ denote the *observed* (reported) state.

The latent three-period history is

$$H = (h_1, h_2, h_3) \in \{0, 1\}^3,$$

and the set of all $2^3 = 8$ possible histories is denoted $\mathcal{H}$. The observed sequence is $s = (s_1, s_2, s_3)$.

A.1.2 True dynamics: first-order Markov chain

The true employment process $\{h_t\}$ is a first-order Markov chain with three parameters:

$$\alpha \equiv \Pr(h_1 = 1), \tag{6}$$

$$\theta_0 \equiv \Pr(h_t = 1 \mid h_{t-1} = 0), \tag{7}$$

$$\theta_1 \equiv \Pr(h_t = 1 \mid h_{t-1} = 1), \tag{8}$$

where θ_0 is the job-finding rate and $1 - \theta_1$ is the separation rate. The initial employment probability α is in general a free parameter; under the **stationarity restriction** (which we impose throughout), the chain is in its ergodic steady state at wave 1, so

$$\alpha = \frac{\theta_0}{\theta_0 + 1 - \theta_1}. \tag{9}$$

The free parameter vector for the Markov dynamics is therefore (θ_0, θ_1), with α derived from (9).

The prior probability of a latent history $h \in \mathcal{H}$ is

$$p(H = h \mid \theta_0, \theta_1) = \alpha^{h_1}(1-\alpha)^{1-h_1} \prod_{t=2}^{3} \Big[\theta_1^{\mathbf{1}\{h_{t-1}=1,\, h_t=1\}} (1-\theta_1)^{\mathbf{1}\{h_{t-1}=1,\, h_t=0\}} \theta_0^{\mathbf{1}\{h_{t-1}=0,\, h_t=1\}} (1-\theta_0)^{\mathbf{1}\{h_{t-1}=0} \tag{10}$$

with α given by (9).

A.1.3 Observation model: state misclassification

Employment status is observed with error. We model misclassification as symmetric, with a single probability $\pi \in [0, \frac{1}{2})$:

$$q_{ab}(\pi) \equiv \Pr(s_t = a \mid h_t = b) = (1-\pi)\,\mathbf{1}\{a = b\} + \pi\,\mathbf{1}\{a \neq b\}, \qquad a, b \in \{0, 1\}. \tag{11}$$

Misclassification is independent across waves and across individuals, conditional on the latent history. The constraint $\pi < \frac{1}{2}$ is a normalisation ensuring that observed states are more often correct than incorrect.

A.1.4 Observed-data likelihood

We observe a sample of N individuals, $i = 1, \ldots, N$, each with survey weight $w_i > 0$ and observed data $y_i = (s_{i1}, s_{i2}, s_{i3})$. Conditional on a history h, the probability of the observed sequence y_i is

$$p(y_i \mid h; \pi) = \prod_{t=1}^{3} q_{s_{it},\, h_t}(\pi).$$

Since the true history is unobserved, the marginal probability of y_i is obtained by summing over all $h \in \mathcal{H}$:

$$p(y_i \mid \Theta) = \sum_{h \in \mathcal{H}} p(H = h \mid \theta_0, \theta_1) \prod_{t=1}^{3} q_{s_{it},\, h_t}(\pi). \tag{12}$$

The observed-data log-likelihood, incorporating survey weights, is

$$\ell(\Theta) = \sum_{i=1}^{N} w_i \log p(y_i \mid \Theta) = \sum_{i=1}^{N} w_i \log \left[\sum_{h \in \mathcal{H}} p(H = h \mid \theta_0, \theta_1) \prod_{t=1}^{3} q_{s_{it}, h_t}(\pi) \right], \tag{13}$$

with parameter vector $\Theta = (\theta_0, \theta_1, \pi)$. This is a finite mixture log-likelihood: each observed sequence is a probability-weighted average over the eight latent histories that could have generated it.

A.2 EM algorithm

Direct numerical maximisation of (13) is complicated by the log-of-sums structure. We develop an EM (Expectation–Maximisation) algorithm that replaces this with a sequence of complete-data maximisations, each of which admits closed-form solutions.

A.2.1 Complete-data formulation

Introduce indicator variables $z_{ih} = \mathbf{1}\{H_i = h\}$ with $\sum_{h \in \mathcal{H}} z_{ih} = 1$. If the latent histories were known, the complete-data likelihood would be

$$L_c(\Theta) = \prod_{i=1}^{N} \prod_{h \in \mathcal{H}} \left[p(H = h \mid \theta_0, \theta_1) \prod_{t=1}^{3} q_{s_{it}, h_t}(\pi) \right]^{w_i z_{ih}}.$$

Taking logarithms:

$$\ell_c(\Theta) = \sum_{i=1}^{N} \sum_{h \in \mathcal{H}} w_i z_{ih} \Bigg\{ \underbrace{\log p(H = h \mid \theta_0, \theta_1)}_{\text{Markov prior}} + \underbrace{\sum_{t=1}^{3} \log q_{s_{it}, h_t}(\pi)}_{\text{misclassification}} \Bigg\}. \tag{14}$$

The complete-data log-likelihood (14) decomposes additively into two blocks: one depending only on (θ_0, θ_1) through the Markov prior, and one depending only on π through the misclassification probabilities. This additive separability is the key structural property that yields closed-form M-step updates.

A.2.2 E-step: responsibilities

Starting from current parameter estimates $\Theta^{(\text{old})} = (\theta_0^{(\text{old})}, \theta_1^{(\text{old})}, \pi^{(\text{old})})$, the E-step computes the posterior probability that individual i has latent history h, conditional on the observed data:

$$\gamma_{ih} \equiv \mathbb{E}[z_{ih} \mid y_i;\, \Theta^{(\text{old})}] = \frac{p(H = h \mid \theta_0^{(\text{old})}, \theta_1^{(\text{old})}) \prod_{t=1}^{3} q_{s_{it},\, h_t}(\pi^{(\text{old})})}{\sum_{h' \in \mathcal{H}} p(H = h' \mid \theta_0^{(\text{old})}, \theta_1^{(\text{old})}) \prod_{t=1}^{3} q_{s_{it},\, h'_t}(\pi^{(\text{old})})}. \quad (15)$$

Each $\gamma_{ih} \in [0,1]$ and $\sum_{h \in \mathcal{H}} \gamma_{ih} = 1$. Intuitively, γ_{ih} is the "responsibility" that history h takes for explaining observation y_i: it combines the prior plausibility of h (from the Markov chain) with how well h accounts for the observed sequence (from the misclassification model).

Substituting (10) and (11) into (15), the numerator for each (i,h) pair is a product of at most six terms (three from the Markov chain, three from misclassification), which can be computed in $O(1)$ time per pair. With N individuals and 8 histories, the full E-step requires $O(N)$ operations.

Sufficient statistics. From the responsibilities, accumulate the following weighted statistics:

$$C_1 = \sum_{i=1}^{N} \sum_{h \in \mathcal{H}} w_i \gamma_{ih}\, h_1, \qquad C_0 = \sum_{i=1}^{N} \sum_{h \in \mathcal{H}} w_i \gamma_{ih}\, (1 - h_1), \quad (16)$$

$$D_1 = \sum_{i=1}^{N} \sum_{h \in \mathcal{H}} w_i \gamma_{ih} \sum_{t=2}^{3} \mathbf{1}\{h_{t-1} = 1\}, \qquad D_0 = \sum_{i=1}^{N} \sum_{h \in \mathcal{H}} w_i \gamma_{ih} \sum_{t=2}^{3} \mathbf{1}\{h_{t-1} = 0\}, \quad (17)$$

$$T_{11} = \sum_{i=1}^{N} \sum_{h \in \mathcal{H}} w_i \gamma_{ih} \sum_{t=2}^{3} \mathbf{1}\{h_{t-1} = 1,\, h_t = 1\}, \quad T_{01} = \sum_{i=1}^{N} \sum_{h \in \mathcal{H}} w_i \gamma_{ih} \sum_{t=2}^{3} \mathbf{1}\{h_{t-1} = 0,\, h_t = 1\}, \quad (18)$$

$$M = \sum_{i=1}^{N} \sum_{h \in \mathcal{H}} w_i \gamma_{ih} \sum_{t=1}^{3} \mathbf{1}\{s_{it} \neq h_t\}. \quad (19)$$

These statistics have natural interpretations: C_1 is the (responsibility-weighted) total mass of individuals truly employed at wave 1; D_1 counts person-periods where the previous state was employment; T_{11} counts employment-to-employment continuations; T_{01} counts nonemployment-to-employment transitions; and M counts misclassified wave-observations.

A.2.3 M-step: parameter updates

The M-step maximises the expected complete-data log-likelihood $Q(\Theta \mid \Theta^{(\text{old})}) \equiv \mathbb{E}[\ell_c(\Theta) \mid \{y_i\}; \Theta^{(\text{old})}]$, obtained by replacing z_{ih} with γ_{ih} in (14). Because the Markov prior block depends only on (θ_0, θ_1) and the misclassification block depends only on π, the two can be maximised independently.

Transition probabilities (closed form). The Markov prior block of Q is

$$\begin{aligned} Q_{\text{prior}}(\theta_0, \theta_1) = \sum_{i,h} w_i \gamma_{ih} \Bigg\{ & h_1 \log \alpha + (1 - h_1) \log(1 - \alpha) \\ & + \sum_{t=2}^{3} \Big[\mathbf{1}\{h_{t-1}{=}1, h_t{=}1\} \log \theta_1 + \mathbf{1}\{h_{t-1}{=}1, h_t{=}0\} \log(1 - \theta_1) \\ & \quad + \mathbf{1}\{h_{t-1}{=}0, h_t{=}1\} \log \theta_0 + \mathbf{1}\{h_{t-1}{=}0, h_t{=}0\} \log(1 - \theta_0) \Big] \Bigg\}. \end{aligned} \tag{20}$$

Under the stationarity restriction (9), the α-dependent terms couple θ_0 and θ_1 through $\alpha(\theta_0, \theta_1)$, so the first-order conditions do not yield fully closed-form expressions. In practice, however, because the contribution of the initial-state term (one observation per individual) is small relative to the transition terms (two transitions per individual), the stationarity coupling has negligible effect on the estimates. We adopt the standard approximation used in the EM literature for short panels: update θ_0 and θ_1 from the transition counts alone, then set α by (9):

$$\hat{\theta}_1 = \frac{T_{11}}{D_1}, \qquad \hat{\theta}_0 = \frac{T_{01}}{D_0}, \qquad \hat{\alpha} = \frac{\hat{\theta}_0}{\hat{\theta}_0 + 1 - \hat{\theta}_1}. \tag{21}$$

In words: $\hat{\theta}_1$ is the fraction of (responsibility-weighted) person-periods starting in employment that result in continued employment, and $\hat{\theta}_0$ is the fraction starting in nonemployment that result in employment. These are the standard Markov chain MLEs, the only difference being that the counts are soft (weighted by responsibilities γ_{ih}) rather than hard.

Misclassification probability (closed form). The misclassification block of Q is

$$Q_{\text{miscl}}(\pi) = \sum_{i,h} w_i \gamma_{ih} \sum_{t=1}^{3} \Big[\mathbf{1}\{s_{it} = h_t\} \log(1-\pi) + \mathbf{1}\{s_{it} \neq h_t\} \log \pi \Big].$$

Differentiating and setting to zero:

$$\hat{\pi} = \frac{M}{3\,W}, \tag{22}$$

where $W = \sum_{i=1}^{N} w_i$ is the total weight. In words: $\hat{\pi}$ is the average (responsibility-weighted) fraction of wave-observations where the observed state disagrees with the latent state. The denominator $3W$ reflects that each of the N individuals contributes three wave-observations.

Summary. The complete M-step consists of three closed-form updates (21)–(22). No numerical optimisation is required.

A.2.4 Iteration and convergence

Starting from an initial guess $\Theta^{(0)} = (\theta_0^{(0)}, \theta_1^{(0)}, \pi^{(0)})$, iterate:

1. **E-step**: compute responsibilities γ_{ih} via (15) and sufficient statistics (16)–(19).
2. **M-step**: update parameters via (21)–(22).

3. **Convergence check**: evaluate the observed-data log-likelihood $\ell(\Theta)$ from (13). Stop when the increase $\ell(\Theta^{(\text{new})}) - \ell(\Theta^{(\text{old})})$ is below a tolerance $\delta > 0$.

The EM algorithm guarantees monotone ascent: $\ell(\Theta^{(\text{new})}) \geq \ell(\Theta^{(\text{old})})$ at every iteration. This follows from the standard EM convergence theorem (Dempster, Laird, and Rubin, 1977; Wu, 1983), since each M-step block independently maximises its portion of Q. In this model, all M-step updates are in closed form, so each iteration is computationally cheap and the ascent property holds exactly (no inner-loop approximation errors).

Because the observed-data likelihood (13) is a finite mixture of products of Bernoulli-type terms, it is bounded above (by zero, since all probabilities are at most one). Combined with monotone ascent, the sequence $\{\ell(\Theta^{(k)})\}$ converges. Under mild regularity conditions (the parameter space is compact and the likelihood is continuous), every limit point of $\{\Theta^{(k)}\}$ is a stationary point of ℓ.

Practical considerations.

- *Multiple starts.* Because the likelihood may have local maxima, we recommend running the algorithm from multiple initial values and retaining the solution with the highest log-likelihood.
- *Boundary constraints.* Enforce $\theta_0, \theta_1 \in [\varepsilon, 1-\varepsilon]$ and $\pi \in [\varepsilon, \frac{1}{2} - \varepsilon]$ for a small $\varepsilon > 0$ to avoid degenerate solutions.
- *Weights.* All sufficient statistics and the log-likelihood use survey weights w_i. The M-step formulas (21)–(22) naturally incorporate weights through the responsibility-weighted counts.

A.2.5 Equivalence to direct numerical maximisation

Because the set of latent histories $\mathcal{H}$ is finite and small ($|\mathcal{H}| = 8$), the observed-data likelihood (13) can be evaluated in closed form: for each observed pattern (s_1, s_2, s_3), the marginal probability (12) is an explicit polynomial in $(\theta_0, \theta_1, \pi)$.

The gradient of $\ell(\Theta)$ can likewise be computed analytically. This permits direct maximisation of (13) via standard gradient-based methods (e.g., BFGS with analytic gradients on a logit-transformed parameter space). The EM algorithm and direct numerical maximisation converge to the same stationary points; the choice between them is one of convenience. We adopt the EM formulation because it generalises naturally when the model is extended to incorporate duration data: the Markov prior and misclassification blocks remain unchanged, and only the emission terms in the E-step and corresponding M-step blocks are added.

A.3 Identification

The model has three free parameters, $\Theta = (\theta_0, \theta_1, \pi)$, and the data provide seven free moments (there are $2^3 = 8$ possible observed sequences (s_1, s_2, s_3), whose probabilities sum to one). The model is therefore overidentified, with four overidentifying restrictions. We now characterise what features of the data identify each parameter.

Observed cell probabilities. Let $n_{abc} = \sum_i w_i \mathbf{1}\{s_{i1} = a,\, s_{i2} = b,\, s_{i3} = c\}$ denote the weighted count of observed pattern $(a, b, c) \in \{0, 1\}^3$, and let $\hat{p}_{abc} = n_{abc}/W$ be the corresponding sample proportion, where $W = \sum_i w_i$. The model-implied probability of each pattern is, from (12),

$$p_{abc}(\Theta) = \sum_{h \in \mathcal{H}} p(H = h \mid \theta_0, \theta_1) \prod_{t=1}^{3} q_{s_t, h_t}(\pi), \tag{23}$$

where $(s_1, s_2, s_3) = (a, b, c)$. Each p_{abc} is a polynomial in $(\theta_0, \theta_1, \pi)$ of degree at most 5 (degree 1 in $\alpha(\theta_0, \theta_1)$, degree 2 from the two transitions, and degree 3 from the three misclassification factors, but with cross-cancellations from the sum over histories).

Identification without misclassification. When $\pi = 0$, observed states equal true states, and the model reduces to a standard first-order Markov chain with two parameters (θ_0, θ_1). These are identified by the observed transition frequencies:

$$\theta_1 = \Pr(s_t = 1 \mid s_{t-1} = 1), \qquad \theta_0 = \Pr(s_t = 1 \mid s_{t-1} = 0),$$

which can be estimated from the four two-wave transition counts. With three waves, there are two independent transition tables (waves 1–2 and 2–3), providing four free moments for two parameters: the model is overidentified and testable (e.g., one can test the time-homogeneity assumption by comparing the two transition tables).

Identification with misclassification. When $\pi > 0$, the observed transition frequencies no longer directly reveal the true transition probabilities. However, the three-wave panel structure provides enough information to disentangle (θ_0, θ_1) from π.

The key insight is that misclassification introduces specific patterns into the observed cell probabilities that cannot be generated by any choice of (θ_0, θ_1) alone. In a true Markov chain (without misclassification), the sequence (s_1, s_3) is conditionally independent of s_1 given s_2 (the Markov property). Misclassification breaks this conditional independence: because s_2 is a noisy signal of h_2, conditioning on s_2 does not fully screen off s_1 from s_3. The degree of residual dependence between s_1 and s_3 given s_2 is a function of π alone (for given transition rates), providing identifying information for the misclassification probability.

More formally, under the model (23), the observed transition matrix from wave 1 to wave 3 (skipping wave 2) satisfies

$$\mathbf{P}_{\text{obs}}^{(1,3)} = \mathbf{Q}(\pi)\, \mathbf{P}^*\, \mathbf{P}^*\, \mathbf{Q}(\pi)^\top, \tag{24}$$

where $\mathbf{P}^*$ is the 2×2 true transition matrix and $\mathbf{Q}(\pi)$ is the 2×2 misclassification matrix with off-diagonal entries π. Meanwhile, the one-step observed transition matrix is $\mathbf{P}_{\text{obs}}^{(1,2)} = \mathbf{Q}(\pi)\, \mathbf{P}^*\, \mathbf{Q}(\pi)^\top$. The relationship between $\mathbf{P}_{\text{obs}}^{(1,3)}$ and $\mathbf{P}_{\text{obs}}^{(1,2)}$ constrains π: for any candidate (θ_0, θ_1), there is at most one value of π that reconciles the one-step and two-step observed transitions.

Counting argument. The model maps three parameters to seven free moments. Generic identification requires the Jacobian of the map $\Theta \mapsto \{p_{abc}(\Theta)\}_{(a,b,c)\neq(0,0,0)}$ to have full column rank 3 at the true parameter value. Analytical computation of this 7×3 Jacobian confirms that it has rank 3 throughout the interior of the parameter space $\{(\theta_0, \theta_1, \pi) : 0 < \theta_0, \theta_1 < 1,\ 0 \leq \pi < \frac{1}{2}\}$, except at isolated degenerate points (e.g., $\theta_0 = 1 - \theta_1$, which makes the chain i.i.d. and collapses the model to two parameters). The model is therefore generically identified.

Overidentification and specification testing. With seven free moments and three parameters, there are four overidentifying restrictions. These can be tested via a likelihood ratio test comparing the saturated model (which fits all seven moments exactly) to the restricted model (23):

$$\mathrm{LR} = 2\big[\ell_{\text{saturated}} - \ell(\hat{\Theta})\big] \ \sim\ \chi^2_4 \quad \text{under } H_0, \tag{25}$$

where the saturated log-likelihood is $\ell_{\text{saturated}} = \sum_{a,b,c} n_{abc} \log \hat{p}_{abc}$ with $\hat{p}_{abc} = n_{abc}/W$. Rejection indicates that the first-order Markov assumption with symmetric misclassification does not adequately describe the observed data, motivating extensions such as higher-order dynamics, asymmetric misclassification, or duration-augmented models.

A.4 Asymmetric misclassification

The symmetric specification (11) assumes that the probability of misclassifying a truly employed individual as nonemployed equals the probability of the reverse error. This assumption can be relaxed by introducing two misclassification parameters:

$$\pi_0 \equiv \Pr(s_t = 1 \mid h_t = 0), \qquad \pi_1 \equiv \Pr(s_t = 0 \mid h_t = 1), \tag{26}$$

where π_0 is the false-positive (spurious employment) rate and π_1 is the false-negative (missed employment) rate. The misclassification probability becomes

$$q_{ab}(\pi_0, \pi_1) = \begin{cases} 1 - \pi_0 & \text{if } a = 0,\, b = 0, \\ \pi_0 & \text{if } a = 1,\, b = 0, \\ \pi_1 & \text{if } a = 0,\, b = 1, \\ 1 - \pi_1 & \text{if } a = 1,\, b = 1. \end{cases}$$

The EM algorithm extends with minimal modification. The E-step (15) uses the asymmetric q_{ab} in place of the symmetric version. The M-step for the transition probabilities (21) is unchanged. The misclassification M-step separates into two updates:

$$\hat{\pi}_0 = \frac{M_0}{\sum_{i,h} w_i \gamma_{ih} \sum_{t=1}^{3} \mathbf{1}\{h_t = 0\}}, \qquad \hat{\pi}_1 = \frac{M_1}{\sum_{i,h} w_i \gamma_{ih} \sum_{t=1}^{3} \mathbf{1}\{h_t = 1\}}, \tag{27}$$

where $M_0 = \sum_{i,h} w_i \gamma_{ih} \sum_t \mathbf{1}\{h_t = 0,\, s_{it} = 1\}$ counts false positives (observed employed when truly nonemployed) and $M_1 = \sum_{i,h} w_i \gamma_{ih} \sum_t \mathbf{1}\{h_t = 1,\, s_{it} = 0\}$ counts false negatives (observed nonemployed when truly employed).

The asymmetric model has four parameters $(\theta_0, \theta_1, \pi_0, \pi_1)$ and the same seven free moments, leaving three overidentifying restrictions. The symmetric restriction $H_0 : \pi_0 = \pi_1$ can be tested via a likelihood ratio test with one degree of freedom.

A.5 Extension I: Observable heterogeneity

A.5.1 Motivation

The baseline model imposes constant transition probabilities (θ_0, θ_1) across all individuals. In practice, employment dynamics vary systematically with observable worker characteristics such as age, education, contract type,[1] and demographics.

[1] Contract type is only observed for individuals who are employed at the time of the survey. For individuals who are nonemployed (or have missing contract information) at wave 1, the

Ignoring this heterogeneity confounds true state persistence with compositional effects, potentially inflating the misclassification estimate. We therefore allow each individual's transition probabilities to depend on a vector of observed covariates.

A.5.2 Covariate specification

Let $x_i \in \mathbb{R}^p$ denote a vector of time-invariant (or wave-one) individual covariates, including an intercept. We replace the scalar transition probabilities with individual-specific probit functions:

$$\theta_1(x_i) \equiv \Pr(h_t = 1 \mid h_{t-1} = 1,\, x_i) = \Phi(\beta_1^\top x_i), \qquad \theta_0(x_i) \equiv \Pr(h_t = 1 \mid h_{t-1} = 0,\, x_i) = \Phi(\beta_0^\top x_i), \tag{28}$$

where Φ is the standard normal CDF and $\beta_0, \beta_1 \in \mathbb{R}^p$ are unknown coefficient vectors. The probit link ensures $\theta_j(x_i) \in (0, 1)$ for all x_i and finite β_j.

Under the stationarity restriction, the individual-specific initial employment probability is

$$\alpha(x_i) = \frac{\theta_0(x_i)}{\theta_0(x_i) + 1 - \theta_1(x_i)}. \tag{29}$$

The prior mass on a history $h \in \mathcal{H}$ is therefore individual-specific:

$$p(H = h \mid \beta_0, \beta_1, x_i) = \alpha(x_i)^{h_1}(1-\alpha(x_i))^{1-h_1} \prod_{t=2}^{3} \Big[\theta_1(x_i)^{\mathbf{1}\{h_{t-1}=1,\, h_t=1\}}\,(1-\theta_1(x_i))^{\mathbf{1}\{h_{t-1}=1,\, h_t=0\}}\,\theta_0(x_i)^{\mathbf{1}\{h_t}$$

The misclassification probability π and the observation model (11) are unchanged.

contract-type variable is set to zero ("no written contract"), so that the corresponding one-hot dummy equals zero for all nonemployed observations.

A.5.3 EM algorithm with covariates

E-step. The responsibility formula (15) carries over with $p(H = h \mid \theta_0, \theta_1)$ replaced by $p(H = h \mid \beta_0, \beta_1, x_i)$:

$$\gamma_{ih} = \frac{p(H = h \mid \beta_0^{(\text{old})}, \beta_1^{(\text{old})}, x_i) \prod_{t=1}^{3} q_{s_{it}, h_t}(\pi^{(\text{old})})}{\sum_{h' \in \mathcal{H}} p(H = h' \mid \beta_0^{(\text{old})}, \beta_1^{(\text{old})}, x_i) \prod_{t=1}^{3} q_{s_{it}, h'_t}(\pi^{(\text{old})})}. \tag{30}$$

M-step: transition coefficients (GEM). With the probit parameterisation (28), the prior block of $Q(\Theta \mid \Theta^{(\text{old})})$ with respect to β_1 is proportional to a *weighted probit log-likelihood.* Adopting the standard short-panel approximation (updating from transition counts, ignoring the initial-state coupling through $\alpha(x_i)$), the objective for β_1 decomposes over person-periods:

$$Q_{\beta_1} = \sum_{i=1}^{N} \sum_{h \in \mathcal{H}} w_i \gamma_{ih} \sum_{t=2}^{3} \mathbf{1}\{h_{t-1} = 1\} \Big[h_t \log \Phi(\beta_1^\top x_i) + (1 - h_t) \log \Phi(-\beta_1^\top x_i) \Big]. \tag{31}$$

This is a probit log-likelihood with pseudo-observations $\{(h_t,\, x_i)\}$ at person-period pairs where $h_{t-1} = 1$, carrying fractional weight $w_i \gamma_{ih}$. Maximising (31) has no closed form.

We therefore apply a **Generalized EM (GEM)** step: rather than finding the exact maximiser of (31), we perform a single iteration of the iteratively reweighted least squares (IRLS) algorithm for weighted probit, which strictly increases Q_{β_1} and hence guarantees monotone ascent of the observed-data log-likelihood. The update for β_0 follows symmetrically from

$$Q_{\beta_0} = \sum_{i,h} w_i \gamma_{ih} \sum_{t=2}^{3} \mathbf{1}\{h_{t-1} = 0\} \Big[h_t \log \Phi(\beta_0^\top x_i) + (1 - h_t) \log \Phi(-\beta_0^\top x_i) \Big]. \tag{32}$$

Under the short-panel approximation Q_{β_0} and Q_{β_1} are separable, so the two probit regressions can be run independently.

M-step: misclassification (closed form). The misclassification block is unchanged from the baseline. Since π does not enter (31)–(32), the update (22) continues to hold exactly:

$$\hat{\pi} = \frac{M}{3W}.$$

Parameter count and identification. The extended parameter vector is $\Theta = (\beta_0, \beta_1, \pi)$, with $2p+1$ free parameters. For each distinct covariate profile x_i, the seven free cell probabilities $p_{abc}(x_i)$ provide identifying information. Identification requires sufficient variation in x_i across the sample. The baseline model is the special case $p = 1$, $\beta_j = (\text{const})$.

A.6 Extension II: Duration dependence

A.6.1 Motivation

The first-order Markov assumption is known to be violated in labour market data due to *duration dependence*: the probability that a spell ends depends on how long it has already lasted. Longer employment spells tend to be more stable (positive duration dependence in retention), and longer unemployment spells may exhibit negative duration dependence (scarring). Ignoring duration dependence can inflate estimated transition rates and, consequently, misclassification estimates. A natural approach is to incorporate reported tenure and time-gap (nonemployment duration) as covariates in the transition probabilities.

A.6.2 Specification

Let $g_{it} \geq 0$ denote reported tenure (employment duration) and $d_{it} \geq 0$ reported time-gap (nonemployment duration) for individual i at wave t, following the notation of `EM tenure epsilon.tex`. We augment the covariate vector from

Section A.5 to include lagged duration:

$$\theta_1(x_{i,t-1}) = \Phi(\beta_1^\top x_{i,t-1}), \qquad\qquad x_{i,t-1} \ni g_{i,t-1}, \tag{33}$$

$$\theta_0(x_{i,t-1}) = \Phi(\beta_0^\top x_{i,t-1}), \qquad\qquad x_{i,t-1} \ni d_{i,t-1}. \tag{34}$$

The transition probability at time t depends on the *lagged* observed duration at $t-1$. The timing is natural: the duration of an employment or nonemployment spell up to wave $t-1$ predicts whether that spell continues or ends by wave t.

A.6.3 The missing-covariate challenge

A fundamental difficulty arises from the interaction between misclassification and the duration variables. Tenure g_{it} is only observed when the *reported* state is employment ($s_{it} = 1$); time-gap d_{it} is only observed when $s_{it} = 0$. In the EM algorithm, however, we must evaluate $p(H = h \mid \beta_0, \beta_1, x_{i,t-1})$ for *every* history $h \in \mathcal{H}$, including histories where the latent state h_{t-1} disagrees with the reported state s_{it-1}.

Concretely, consider the transition term at $t = 2$ for a history with $h_1 = 1$ (truly employed at wave 1). The corresponding transition probability is $\theta_1(x_{i1})$, which requires tenure g_{i1} as a covariate. But if $s_{i1} = 0$ (the individual is misclassified as nonemployed at wave 1), we observe time-gap d_{i1} instead, and tenure g_{i1} is missing for that individual-wave. By symmetry, for a history with $h_1 = 0$ but $s_{i1} = 1$, the required time-gap d_{i1} is not observed.

This creates a *missing-covariate-under-misclassification* problem: the covariates required to evaluate the prior for a given history depend on that history, but some covariates are only available for histories consistent with the observed states. The problem is not a simple missing- data problem because the missingness mechanism itself depends on the latent variable being integrated over. Standard imputation approaches would require modelling the joint distribution of duration and employment status — that is, effectively re-introducing the full duration-emission model of the kind developed in `EM tenure.tex`.

This challenge is left for further development. Any practical implementation of duration dependence as a transition-rate covariate must resolve the treatment of duration values when the observation and the latent history disagree.

A.7 Extension III: Unobserved heterogeneity

A.7.1 Motivation

Observable covariates (Section A.5) capture systematic variation in employment dynamics. However, persistent unobserved traits — such as motivation, ability, or family background — may also drive heterogeneity in transition rates. If workers who are inherently more employment-stable are observationally identical to less-stable workers, the pooled first-order Markov model mis-specifies the data-generating process, and estimated transition rates conflate genuine state-switching with compositional turnover. This could again inflate misclassification estimates. We address this using a **finite mixture model (FMM)** with two latent worker types.

A.7.2 Two-type FMM specification

Suppose the population consists of two latent types, $k \in \{A, B\}$, with mixing weight $\phi \in (0, 1)$:

$$\Pr(K_i = A) = \phi, \qquad \Pr(K_i = B) = 1 - \phi.$$

Each type has its own true employment transition probabilities, (θ_0^A, θ_1^A) and (θ_0^B, θ_1^B), and its own ergodic employment rate:

$$\alpha^k = \frac{\theta_0^k}{\theta_0^k + 1 - \theta_1^k}, \qquad k \in \{A, B\}. \tag{35}$$

The misclassification probability π is *shared* across types: it reflects the reliability of the survey instrument rather than any individual trait. The complete parameter vector is $\Theta = (\theta_0^A, \theta_1^A, \theta_0^B, \theta_1^B, \phi, \pi)$.

Observed-data likelihood. The marginal probability of observation y_i integrates over both the latent history $h \in \mathcal{H}$ and the latent type k:

$$p(y_i \mid \Theta) = \sum_{k \in \{A,B\}} \Pr(K_i = k) \sum_{h \in \mathcal{H}} p(H = h \mid \theta_0^k, \theta_1^k) \prod_{t=1}^{3} q_{s_{it},\, h_t}(\pi). \tag{36}$$

The latent space has expanded from $|\mathcal{H}| = 8$ histories to $|\mathcal{H}| \times 2 = 16$ history–type pairs.

A.7.3 EM algorithm for the FMM

Complete-data formulation. Introduce indicators $z_{ihk} = \mathbf{1}\{H_i = h,\ K_i = k\}$ with $\sum_{h,k} z_{ihk} = 1$. The complete-data log-likelihood is

$$\ell_c(\Theta) = \sum_{i=1}^{N} \sum_{k} \sum_{h \in \mathcal{H}} w_i z_{ihk} \Big\{ \log \Pr(K_i = k) + \underbrace{\log p(H = h \mid \theta_0^k, \theta_1^k)}_{\text{type-}k\text{ Markov prior}} + \underbrace{\sum_{t=1}^{3} \log q_{s_{it},\, h_t}(\pi)}_{\text{misclassification}} \Big\}. \tag{37}$$

The three blocks are additively separable: the type-weight block depends only on ϕ, the type-k Markov prior block depends only on (θ_0^k, θ_1^k), and the misclassification block depends only on π. All blocks can be maximised independently.

E-step: joint responsibilities. Starting from current parameter estimates $\Theta^{(\text{old})}$, the E-step computes the posterior probability that individual i has history h *and* is type k:

$$\gamma_{ihk} \equiv \mathbb{E}[z_{ihk} \mid y_i;\, \Theta^{(\text{old})}] = \frac{\Pr(K_i = k)\, p(H = h \mid \theta_0^{k(\text{old})}, \theta_1^{k(\text{old})}) \prod_{t=1}^{3} q_{s_{it},\, h_t}(\pi^{(\text{old})})}{p(y_i \mid \Theta^{(\text{old})})}, \tag{38}$$

where the denominator is (36) evaluated at $\Theta^{(\text{old})}$. Each $\gamma_{ihk} \geq 0$ and $\sum_{h,k} \gamma_{ihk} = 1$. The type-marginalised responsibility $\gamma_{ik} = \sum_h \gamma_{ihk}$ is the posterior probability that individual i belongs to type k.

M-step: type weights (closed form). Maximising the type-weight block of Q yields

$$\hat{\phi} = \frac{\sum_{i=1}^{N} w_i \gamma_{iA}}{W}, \tag{39}$$

where $W = \sum_i w_i$. This is the responsibility-weighted fraction of individuals assigned to type A.

M-step: type-specific transition probabilities (closed form). For each type $k \in \{A, B\}$, define type-specific sufficient statistics:

$$T_{11}^k = \sum_{i=1}^{N} \sum_{h \in \mathcal{H}} w_i \gamma_{ihk} \sum_{t=2}^{3} \mathbf{1}\{h_{t-1} = 1,\, h_t = 1\}, \qquad D_1^k = \sum_{i,h} w_i \gamma_{ihk} \sum_{t=2}^{3} \mathbf{1}\{h_{t-1} = 1\}, \tag{40}$$

$$T_{01}^k = \sum_{i=1}^{N} \sum_{h \in \mathcal{H}} w_i \gamma_{ihk} \sum_{t=2}^{3} \mathbf{1}\{h_{t-1} = 0,\, h_t = 1\}, \qquad D_0^k = \sum_{i,h} w_i \gamma_{ihk} \sum_{t=2}^{3} \mathbf{1}\{h_{t-1} = 0\}. \tag{41}$$

The M-step updates are the same ratio formulas as the baseline, now applied type by type:

$$\hat{\theta}_1^k = \frac{T_{11}^k}{D_1^k}, \qquad \hat{\theta}_0^k = \frac{T_{01}^k}{D_0^k}, \qquad \hat{\alpha}^k = \frac{\hat{\theta}_0^k}{\hat{\theta}_0^k + 1 - \hat{\theta}_1^k}. \tag{42}$$

The two types are updated independently: there is no cross-type coupling in the M-step.

M-step: misclassification (closed form, pooled). Since π is shared across types, the misclassification update pools over both types and all histories:

$$\hat{\pi} = \frac{M}{3W}, \qquad M = \sum_{i=1}^{N} \sum_{k} \sum_{h \in \mathcal{H}} w_i \gamma_{ihk} \sum_{t=1}^{3} \mathbf{1}\{s_{it} \neq h_t\}. \tag{43}$$

The structure is identical to (22); only the definition of M now marginalises over types.

Parameter count and identification. The FMM adds four parameters relative to the baseline: θ_0^B, θ_1^B (the second type's transitions) and ϕ (the mixing weight), while θ_0^A, θ_1^A correspond to the baseline θ_0, θ_1. With five parameters $(\theta_0^A, \theta_1^A, \theta_0^B, \theta_1^B, \phi)$ plus π against seven free moments, the model is just-identified for the Markov parameters and requires care. In practice, identification of two-component mixtures in short Markov chain panels relies on the different implied patterns of serial dependence across the eight cell probabilities; separability of the two types is aided by the three-wave structure. Multiple starts are strongly recommended, as the FMM likelihood typically has local maxima corresponding to label switching between types A and B.

A.8 Extension IV: Inconsistency-augmented misclassification

A.8.1 Motivation

Identification in the baseline model rests on the violation of the conditional Markov property caused by misclassification. An independent source of evidence for misclassification is the existence of respondents who exhibit obviously erroneous answers to other survey questions. Deterministic variables — such as age, which can only remain constant or increase by exactly one year between quarterly waves, or education level, which is non-decreasing and constrained in how rapidly it can advance — provide a clean signal of unreliable survey responses.

If a respondent misreports age or education, they are plausibly more likely to have also misreported their employment status. Incorporating inconsistency indicators into the misclassification probability therefore (i) increases the model's ability to identify π, and (ii) provides a direct empirical test of the misclassification hypothesis: a significant positive coefficient on an inconsistency indicator means that workers who demonstrably misreport one variable also exhibit employment transition patterns consistent with misclassification error.

A.8.2 Inconsistency indicators

Let a_{it} denote reported age and e_{it} reported education level at wave t. We define wave-specific inconsistency indicators by attributing each pairwise discrepancy to one of the three waves.

Age inconsistencies. Between consecutive waves, age should either remain constant or increase by exactly one unit. Define pairwise age-gap dummies:

$$V^{\text{age}}_{i,t,t+1} \equiv \mathbf{1}\Big\{a_{i,t+1} - a_{it} \notin \{0,1\}\Big\}, \quad t \in \{1,2\}.$$

Under the assumption that age is measured with error in at most one wave, the wave-attribution rule is:

$$Y^{\text{age}}_{i1} = V^{\text{age}}_{i12}\,(1 - V^{\text{age}}_{i23}), \quad Y^{\text{age}}_{i2} = V^{\text{age}}_{i12}\,V^{\text{age}}_{i23}, \quad Y^{\text{age}}_{i3} = (1 - V^{\text{age}}_{i12})\,V^{\text{age}}_{i23}. \tag{44}$$

An inconsistency in waves 1–2 but not 2–3 is attributed to wave 1; an inconsistency in waves 2–3 but not 1–2 is attributed to wave 3; a simultaneous inconsistency in both pairs is attributed to wave 2, since a single erroneous wave 2 response generates both pairwise discrepancies.

Education inconsistencies. Education level should be non-decreasing across waves and cannot jump by more than one educational category in a single quarter. Define:

$$V^{\text{edu}}_{i,t,t+1} \equiv \mathbf{1}\Big\{e_{i,t+1} < e_{it} \text{ or } e_{i,t+1} - e_{it} > 1\Big\}, \quad t \in \{1,2\},$$

with wave attribution Y^{edu}_{it} constructed analogously to (44).

A.8.3 Extended misclassification model

We model the wave-specific misclassification probability as

$$\pi_{it} = \frac{1}{2}\,\sigma\Big(\delta_0 + \delta_1 Y^{\text{age}}_{it} + \delta_2 Y^{\text{edu}}_{it}\Big), \tag{45}$$

where $\sigma(u) = (1 + e^{-u})^{-1}$ is the logistic function. Since $\sigma(\cdot) \in (0, 1)$, the factor $\frac{1}{2}$ ensures $\pi_{it} \in (0, \frac{1}{2})$ for all values of the linear predictor, preserving the identifiability constraint. The parameter vector for the misclassification block is $\delta = (\delta_0, \delta_1, \delta_2)$.

Baseline parameters. The baseline misclassification probability (in the absence of any inconsistency) is $\pi_{it}^{\text{base}} = \frac{1}{2}\,\sigma(\delta_0)$. This corresponds to the $\hat{\pi}$ of the symmetric baseline model and is recovered by setting $\delta_1 = \delta_2 = 0$.

A.8.4 EM algorithm with inconsistency indicators

E-step. The responsibility formula (15) is modified by replacing the scalar π with the individual- and wave-specific π_{it} from (45):

$$\gamma_{ih} = \frac{p(H = h \mid \theta_0, \theta_1)\, \prod_{t=1}^{3} q_{s_{it},\, h_t}(\pi_{it}^{(\text{old})})}{\sum_{h' \in \mathcal{H}} p(H = h' \mid \theta_0, \theta_1)\, \prod_{t=1}^{3} q_{s_{it},\, h'_t}(\pi_{it}^{(\text{old})})}. \tag{46}$$

The Markov prior and transition-probability blocks are unchanged.

M-step: misclassification coefficients (GEM). The expected complete-data log-likelihood with respect to δ is

$$Q_{\text{miscl}}(\delta) = \sum_{i=1}^{N} \sum_{h \in \mathcal{H}} w_i \gamma_{ih} \sum_{t=1}^{3} \Big[\mathbf{1}\{s_{it} = h_t\} \log(1 - \pi_{it}) + \mathbf{1}\{s_{it} \neq h_t\} \log \pi_{it} \Big], \tag{47}$$

where π_{it} depends on δ via (45). The gradient of (47) with respect to δ is

$$\nabla_\delta Q_{\text{miscl}} = \sum_{i,h} w_i \gamma_{ih} \sum_{t=1}^{3} \frac{\partial \pi_{it}}{\partial \delta} \left[\frac{\mathbf{1}\{s_{it} \neq h_t\}}{\pi_{it}} - \frac{\mathbf{1}\{s_{it} = h_t\}}{1 - \pi_{it}} \right], \tag{48}$$

where $\partial \pi_{it} / \partial \delta = \frac{1}{2}\,\sigma(\cdot)(1 - \sigma(\cdot))\, f_{it}$ with $f_{it} = (1,\, Y_{it}^{\text{age}},\, Y_{it}^{\text{edu}})^\top$. A GEM step performs one Newton–Raphson update on (47) using gradient (48) and the analytic Hessian, which is negative semi-definite and therefore guarantees a strict increase in Q_{miscl} provided the step is bounded appropriately.

M-step: transition probabilities (closed form). The transition-probability block is unchanged from the baseline. The M-step updates (21) continue to hold exactly, since the Markov prior is independent of δ.

Interpretation. A positive and significant estimate $\hat{\delta}_1 > 0$ means that waves with an age inconsistency exhibit employment patterns consistent with elevated misclassification: the responsibilities γ_{ih} are systematically allocated toward histories where the observed state s_{it} disagrees with the true state h_t. This constitutes independent, auxiliary evidence for the misclassification hypothesis — evidence that does not rely solely on violations of the Markov property.

B AR(2) extension: full derivation

B.1 Motivation

The baseline specification models true employment as a first-order Markov chain: the transition probability at wave t depends only on h_{t-1}. This rules out duration dependence—the empirically important phenomenon whereby the probability of exiting a state depends on how long the individual has been in that state. In discrete time with quarterly observations, the minimal extension that captures one-period duration dependence is a second-order Markov chain, where $\Pr(h_t \mid h_{t-1}, h_{t-2})$ depends on the two most recent states. Testing for second-order dependence is equivalent to testing whether the additional parameters are zero, providing a natural specification test for the baseline model.

This section extends the baseline EM algorithm to the AR(2) case. The observation model (symmetric misclassification with probability π, equation (11) of the baseline) is unchanged. The panel is extended from three to four waves to provide sufficient transitions for identification.

B.2 Model

B.2.1 States and notation

Consider a population observed at four equally spaced waves $t \in \{1, 2, 3, 4\}$, separated by quarterly intervals of $\Delta = 0.25$ years. As in the baseline, $h_t \in \{0, 1\}$ denotes the true (latent) employment state and $s_t \in \{0, 1\}$ the observed state.

The latent four-period history is

$$H = (h_1, h_2, h_3, h_4) \in \{0, 1\}^4,$$

and the set of all $2^4 = 16$ possible histories is denoted $\mathcal{H}$. The observed sequence is $s = (s_1, s_2, s_3, s_4)$.

B.2.2 True dynamics: second-order Markov chain

The true employment process $\{h_t\}$ is a second-order Markov chain. The transition probabilities are parameterised as

$$\Pr(h_t = 1 \mid h_{t-1}, h_{t-2}) = \begin{cases} \theta_0 & \text{if } h_{t-2} = 0,\ h_{t-1} = 0, \\ \theta_0 + \theta_{01} & \text{if } h_{t-2} = 1,\ h_{t-1} = 0, \\ 1 - \theta_1 - \theta_{10} & \text{if } h_{t-2} = 0,\ h_{t-1} = 1, \\ 1 - \theta_1 & \text{if } h_{t-2} = 1,\ h_{t-1} = 1, \end{cases} \tag{49}$$

where $\theta_0 \in (0, 1)$ is the baseline job-finding rate (from two consecutive periods of nonemployment), $\theta_1 \in (0, 1)$ is the baseline separation rate (from two consecutive periods of employment), and the *duration-dependence increments* θ_{01} and θ_{10} capture the effect of the state two periods ago. When $\theta_{01} = \theta_{10} = 0$, the model reduces to the AR(1) baseline. All four transition probabilities are constrained to lie in $(0, 1)$, which imposes $\theta_0 + \theta_{01} \in (0, 1)$ and $\theta_1 + \theta_{10} \in (0, 1)$.

For notational convenience, define the four unrestricted transition probabilities:

$$p_{00} \equiv \theta_0, \qquad p_{10} \equiv \theta_0 + \theta_{01}, \qquad p_{01} \equiv 1 - \theta_1 - \theta_{10}, \qquad p_{11} \equiv 1 - \theta_1, \tag{50}$$

where $p_{jk} = \Pr(h_t = 1 \mid h_{t-2} = j,\, h_{t-1} = k)$. The inverse mapping recovers the structural parameters:

$$\theta_0 = p_{00}, \qquad \theta_{01} = p_{10} - p_{00}, \qquad \theta_1 = 1 - p_{11}, \qquad \theta_{10} = p_{11} - p_{01}. \tag{51}$$

B.2.3 Stationary distribution

Under stationarity, the chain is in its ergodic steady state at waves 1 and 2. The second-order chain on $\{h_t\}$ is equivalent to a first-order chain on the state pair $(h_{t-1}, h_t) \in \{0,1\}^2$, with 4×4 transition matrix. The stationary distribution $\alpha(h_1, h_2) \equiv \Pr(h_1, h_2)$ satisfies the balance equations.

Define the normalising constant

$$\Phi \equiv \theta_0(\theta_{10} - 1) + \theta_1(\theta_{01} - 1). \tag{52}$$

Then the joint stationary probabilities are

$$\alpha(0,0) = \frac{\theta_1\,(\theta_0 + \theta_{01} - 1)}{\Phi}, \tag{53}$$

$$\alpha(1,0) = \frac{-\theta_0\,\theta_1}{\Phi}, \tag{54}$$

$$\alpha(0,1) = \frac{-\theta_0\,\theta_1}{\Phi}, \tag{55}$$

$$\alpha(1,1) = \frac{\theta_0\,(\theta_1 + \theta_{10} - 1)}{\Phi}. \tag{56}$$

Note that $\alpha(1,0) = \alpha(0,1)$, which follows from the detailed balance of the embedded pair chain. The marginal employment rate at any single wave is $\alpha(0,1) + \alpha(1,1)$.

B.2.4 Prior probability of a latent history

The prior probability of a history $h = (h_1, h_2, h_3, h_4)$ is

$$p(H = h \mid \Theta) = \alpha(h_1, h_2) \prod_{t=3}^{4} p_{h_{t-2}, h_{t-1}}^{\mathbf{1}\{h_t=1\}} (1 - p_{h_{t-2}, h_{t-1}})^{\mathbf{1}\{h_t=0\}}, \tag{57}$$

where $\alpha(h_1, h_2)$ is given by (53)–(56) and p_{jk} by (50). The product runs over waves 3 and 4, each contributing one second-order transition factor.

B.2.5 Observed-data likelihood

The observation model is identical to the baseline: symmetric misclassification with probability π (equation (11) of the baseline), independent across waves conditional on the latent history. The marginal probability of the observed sequence $y_i = (s_{i1}, s_{i2}, s_{i3}, s_{i4})$ is

$$p(y_i \mid \Theta) = \sum_{h \in \mathcal{H}} p(H = h \mid \Theta) \prod_{t=1}^{4} q_{s_{it}, h_t}(\pi), \tag{58}$$

and the observed-data log-likelihood is

$$\ell(\Theta) = \sum_{i=1}^{N} w_i \log p(y_i \mid \Theta), \tag{59}$$

with parameter vector $\Theta = (\theta_0, \theta_{01}, \theta_{10}, \theta_1, \pi)$. This is a finite mixture over the 16 latent histories.

B.3 EM algorithm

The EM algorithm follows the same structure as the baseline. The complete-data log-likelihood decomposes additively into a *dynamics block* (depending on the transition parameters through the AR(2) prior) and a *misclassification block* (depending only on π), enabling separate maximisation in the M-step.

B.3.1 E-step: responsibilities

Starting from current estimates $\Theta^{(\mathrm{old})}$, the E-step computes posterior responsibilities

$$\gamma_{ih} = \frac{p(H = h \mid \Theta^{(\mathrm{old})}) \prod_{t=1}^{4} q_{s_{it},\, h_t}(\pi^{(\mathrm{old})})}{\sum_{h' \in \mathcal{H}} p(H = h' \mid \Theta^{(\mathrm{old})}) \prod_{t=1}^{4} q_{s_{it},\, h'_t}(\pi^{(\mathrm{old})})}, \tag{60}$$

for each individual i and history $h \in \mathcal{H}$. Each numerator is a product of at most nine terms: four from the AR(2) prior (one joint initial probability and two transition factors) and four from misclassification. With N individuals and 16 histories, the E-step requires $O(N)$ operations.

Sufficient statistics. From the responsibilities, accumulate the following weighted counts. For the transition parameters, define four origin-type counters and four transition counters, indexed by the state pair (h_{t-2}, h_{t-1}):

$$D_{jk} = \sum_{i=1}^{N} \sum_{h \in \mathcal{H}} w_i \gamma_{ih} \sum_{t=3}^{4} \mathbf{1}\{h_{t-2} = j,\, h_{t-1} = k\}, \tag{61}$$

$$T_{jk} = \sum_{i=1}^{N} \sum_{h \in \mathcal{H}} w_i \gamma_{ih} \sum_{t=3}^{4} \mathbf{1}\{h_{t-2} = j,\, h_{t-1} = k,\, h_t = 1\}, \tag{62}$$

for $j, k \in \{0, 1\}$. Here D_{jk} counts (responsibility-weighted) person-periods where the two preceding states were (j, k), and T_{jk} counts the subset of those that transition to employment.

For the misclassification parameter:

$$M = \sum_{i=1}^{N} \sum_{h \in \mathcal{H}} w_i \gamma_{ih} \sum_{t=1}^{4} \mathbf{1}\{s_{it} \neq h_t\}. \tag{63}$$

B.3.2 M-step: parameter updates

The M-step maximises $Q(\Theta \mid \Theta^{(\mathrm{old})})$ by updating each block independently.

Transition probabilities (closed form). As in the baseline, we adopt the standard approximation for short panels: update the transition parameters from the transition counts alone, then set the initial distribution by stationarity. The unrestricted transition probabilities have closed-form updates:

$$\hat{p}_{jk} = \frac{T_{jk}}{D_{jk}}, \qquad j,k \in \{0,1\}. \tag{64}$$

Each $\hat{p}_{jk}$ is the responsibility-weighted fraction of person-periods with origin pair (j,k) that result in employment. The structural parameters are then recovered via (51):

$$\hat{\theta}_0 = \hat{p}_{00}, \qquad \hat{\theta}_{01} = \hat{p}_{10} - \hat{p}_{00}, \qquad \hat{\theta}_1 = 1 - \hat{p}_{11}, \qquad \hat{\theta}_{10} = \hat{p}_{11} - \hat{p}_{01}. \tag{65}$$

The stationary distribution $\alpha(h_1,h_2)$ is updated from the new $\hat{\theta}$ values via (52)–(56).

Misclassification probability (closed form).

$$\hat{\pi} = \frac{M}{4\,W}, \tag{66}$$

where $W = \sum_{i=1}^{N} w_i$. The denominator is $4W$ (rather than $3W$ in the baseline) because each individual now contributes four wave-observations.

Summary. The complete M-step consists of five closed-form updates: four transition probabilities (64) and one misclassification probability (66). No numerical optimisation is required.

B.3.3 Iteration and convergence

The iteration scheme and convergence properties are identical to the baseline (Section A.2.4 of the baseline). The EM guarantees monotone ascent of $\ell(\Theta)$; all M-step updates are in closed form; the observed-data likelihood is bounded above; and every limit point of the parameter sequence is a stationary point.

The same practical recommendations apply: multiple starts to guard against local maxima, boundary constraints $p_{jk} \in [\varepsilon, 1-\varepsilon]$ and $\pi \in [\varepsilon, \frac{1}{2}-\varepsilon]$, and survey weights throughout.

B.3.4 Testing for second-order dependence

The AR(1) model is nested within the AR(2) model under the restriction $\theta_{01} = \theta_{10} = 0$, or equivalently $p_{10} = p_{00}$ and $p_{01} = p_{11}$. This can be tested via a likelihood ratio test:

$$\mathrm{LR} = 2\big[\ell(\hat{\Theta}_{\mathrm{AR}(2)}) - \ell(\hat{\Theta}_{\mathrm{AR}(1)})\big] \;\sim\; \chi^2_2 \quad \text{under } H_0 : \theta_{01} = \theta_{10} = 0. \tag{67}$$

B.4 Identification

The model has five free parameters, $\Theta = (\theta_0, \theta_{01}, \theta_{10}, \theta_1, \pi)$. The data provide $2^4 - 1 = 15$ free moments (the probabilities of the 16 observed patterns (s_1, s_2, s_3, s_4), summing to one). The model is therefore overidentified with ten overidentifying restrictions.

Identification without misclassification. When $\pi = 0$, observed states equal true states, and the four transition probabilities p_{jk} are directly identified from the observed second-order transition frequencies:

$$p_{jk} = \Pr(s_t = 1 \mid s_{t-2} = j,\ s_{t-1} = k),$$

which can be estimated from the eight second-order transition counts across waves 2–3 and 3–4. With 15 free moments and four parameters, the model without misclassification has 11 overidentifying restrictions.

Identification with misclassification. When $\pi > 0$, the key identifying argument parallels the baseline. In a true second-order Markov chain, h_4 is conditionally independent of h_1 given (h_2, h_3). Misclassification breaks this conditional independence in the observed data: conditioning on (s_2, s_3) does not fully screen off s_1 from s_4 because each s_t is a noisy signal of h_t. The residual dependence between s_1 and s_4 given (s_2, s_3) provides identifying information for π.

More formally, the model maps five parameters to 15 free moments. The Jacobian of the map $\Theta \mapsto \{p_{abcd}(\Theta)\}$ has full column rank 5 throughout the interior of the parameter space (verified analytically via symbolic computation), confirming generic identification. The ten overidentifying restrictions can be tested via a likelihood ratio test against the saturated model:

$$\mathrm{LR} = 2\big[\ell_{\text{saturated}} - \ell(\hat{\Theta})\big] \;\sim\; \chi^2_{10} \quad \text{under } H_0. \tag{68}$$

Rejection indicates that the second-order Markov assumption with symmetric misclassification does not adequately describe the data, motivating further extensions such as the tenure-augmented contamination model.

C Tenure contamination model: full derivation

C.1 Motivation

The base EM-tenure specification (`EM tenure.tex`) relies on a Gaussian increment density $N(0, 2\sigma_g^2)$ for the tenure-measurement component, an EMG $f_{\text{EMG}}(\cdot; \lambda_g, \sigma_g^2)$ for spell levels, and a single error parameter π.[2] Three empirical facts in the QLFS rule out the Gaussian measurement model:

1. **Tenure is integer-month, not continuous Gaussian.** Reported tenures are integer-month value which does not concur with the $\mathcal{N}(0, \sigma_g^2)$ noise assumption.
2. **63.2% of E→E continuation increments are exactly 0.25 years**. The Gaussian MLE for σ_g^2 collapses to zero. Among the remaining 36.8%:

[2](The duration-contamination extension `EM tenure rho.tex`) also relies on a Gaussian increment density, and relies on π and ρ

1.5% are within 0.5 months of 3, 7% are within 1–6 months, and 28% are wildly off (negatives, year-shifted, or large positive deviations). This is a so-called *contamination distribution* —i.e. a mixture —not a Gaussian additive error.

3. **36.7% of within-panel, new job entrants (state pattern $s_{t-1} = 0, s_t = 1$) report tenure > 12 months.** A genuine job entry cannot have 12+ months of tenure. These observations must be misclassifications at $t-1$: the respondent was actually employed at $t-1$ but misclassified, and the tenure at wave t reflects the ongoing spell (or the tenure is wrong).

Fact (3) is an important justification for this ε tenure model: tenure data corroborates employment-state Markov violations in identifying misclassification. The base endogenous EM tenure model fails to exploit this corroboration fully. The base EM tenure model collapses $\sigma_g^2 \to 0$ and routes all contamination through π (which inflates to over 25%).[3] It models the wave-3 tenure under the latent history $h = (1, 1, 1)$ with intermediate misclassification as an EMG draw *independent* of g_1, discarding the structural information that waves 1 and 3 belong to the same spell.

Importantly, in the base EM tenure model, random EMG draws for tenure can only occur when there has truly been a misclassification of employment status. However, we note that it is possible that employment status is reliable even when tenure is not. Moreover, even if an individual is completely mismatched such that tenure becomes a random draw from the tenure distribution, fthere would still be an approximately 50% chance that their employment status is correct by coincidence due to the employment rate being near 50%. Under the base EM tenure model, it is required that for these observations true employment be different to observed employment, and to accommodate that it drives up both mislclassification and true transitions. For example, if $h = (1, 1, 1)$ and $s = (1, 1, 1)$ but $g = (1, 5, 1.5)$, the base EM tenure model would see that tenure in wave 2 is inconsistent with 0.25 year increments, and the would not explain

[3]The ρ model restores plausible $\pi \approx 4\%$ but absorbs all tenure inconsistency into ρ, so tenure does not actively inform π beyond what state-Markov violations alone provide.

this tenure pattern using σ because it has collapsed to zero as discussed above. The only other way to account for this tenure is to declare a misclassification at wave 2, which requires thinking that $h = (1, 0, 1)$ and $s = (1, 1, 1)$ such that there is both a transition into and out of true employment, but that there was a misclassification at wave 2. In this way, both the true transitions and the misclassification estimates are driven up.

This contamination specification addresses all three issues. It replaces the Gaussian + EMG measurement model with a point-mass + Exponential contamination mixture, eliminating σ_g^2 entirely, and introduces a *spell-pair* joint emission that uses tenure flanks to discriminate misclassification from genuine transition.

C.2 Notation and terminology

We follow `EM tenure.tex` notation. Waves $t \in \{1, 2, 3\}$, latent state $h_t \in \{0, 1\}$, observed state $s_t \in \{0, 1\}$, tenure $g_t \geq 0$, nonemployment duration d_t (interval-censored as a category $k_t \in \{1, \ldots, 7\}$). The latent 3-period history is $H = (h_1, h_2, h_3)$ and the set of histories $\mathcal{H}$ has 8 elements. The clock spacing is $\Delta = 0.25$ years. Parameters: $\Theta = (\alpha, \theta_0, \theta_1, \pi, \varepsilon, \lambda_g, \lambda_d)$. When we move to estimation (Section 4), we index individuals $i = 1, \ldots, N$ with survey weights w_i and write s_{it}, g_{it}, etc. for individual-level observations.

A **clock** (or *deterministic clock*) refers to the latent tenure or nonemployment duration that would be observed if the true state history h were known and there were no measurement error. Given h, the clock evolves deterministically on the quarterly grid: each continuation wave increments the clock by $\Delta = 0.25$ years, while a state transition resets it. For example, three consecutive waves of true employment produce the tenure clock path $g_1^*,\ g_1^* + 0.25,\ g_1^* + 0.50$. Observed tenure may deviate from the clock due to contamination (see below).

A **flank** is a tenure observation at one end of a latent employment spell that spans an intermediate wave where tenure is unobserved (because $s_t = 0$). For example, under state pattern $s = (1, 0, 1)$ and latent history $h = (1, 1, 1)$, the spell runs $[1, 3]$ but tenure is observed only at waves 1 and 3; g_1 is the *left flank* and

g_3 the *right flank*. If both flanks are clock-consistent, they satisfy $g_3 = g_1 + 2\Delta$. The spell-pair emission (Section C.3.4) exploits this structural link to identify contamination and, indirectly, misclassification.

A **branch** refers to one component of a mixture density. Throughout this document, the tenure measurement model is a two-component mixture governed by the latent contamination indicator m_t: the *clock-consistent branch* ($m_t = 0$, probability $1-\varepsilon$) assigns the observation to its deterministic clock value (a point mass, i.e. a point with probability 1), while the *contaminated branch* ($m_t = 1$, probability ε) treats the observation as an independent $\text{Exp}(\lambda_g)$ draw. At the spell level, the per-wave branches combine multiplicatively across the K observed waves, so the spell emission (71) sums over all 2^K contamination patterns. Throughout this document, a wave is called **clean** when $m_t = 0$ (its tenure equals the clock) and **contaminated** when $m_t = 1$ (its tenure is an independent draw unrelated to the clock).

C.3 Tenure measurement model

C.3.1 Spell-length distribution: Exponential

We model the latent employment-spell length as $T_g \sim \text{Exp}(\lambda_g)$ and the latent nonemployment-spell length as $T_d \sim \text{Exp}(\lambda_d)$, without an additive Gaussian noise component. The motivation is twofold:

- Identification of (λ, σ^2) jointly in an EMG is delicate and is empirically undermined here by the point-mass at zero increment. Removing σ^2 makes λ identification immediate from the empirical mean of the spell distribution.
- The remaining tenure variation (deviations from clock-consistency) is captured by the contamination component ε, described below. There is no role left for an additive Gaussian.

C.3.2 Contamination model: point-mass + Exponential

Let $m_t \in \{0, 1\}$ be a latent indicator for whether the wave-t tenure report is *clock-consistent* ($m_t = 0$, probability $1 - \varepsilon$) or *contaminated* ($m_t = 1$, probability ε). Contamination represents recall error, year-rounding, dwelling-respondent substitution, or any mechanism that produces a tenure value unrelated to the clock.

The clock-consistent branch reports the deterministic clock value exactly (point mass). The contaminated branch reports an independent draw from the spell-length distribution $\text{Exp}(\lambda_g)$:

$$g_t \mid h, m_t = 0,\, s_t = 1 \;=\; g_t^*(h) \quad \text{(deterministic)}, \qquad g_t \mid h, m_t = 1,\, s_t = 1 \;\sim\; \text{Exp}(\lambda_g), \tag{69}$$

where $g_t^*(h)$ is the latent clock value defined below in (70). This measurement model applies only at waves where tenure is observed ($s_t = 1$); when $s_t = 0$ the survey records a timegap category instead and the tenure emission is not evaluated.

The contamination indicators are independent across waves, so the joint probability of all the tenure observations within a spell is just the product of each wave's individual contribution.

C.3.3 Latent clock and spell-pair structure

For a maximal latent E-spell spanning waves $a \le t \le b$ (i.e., $h_t = 1$ for all $a \le t \le b$), the latent tenure path is

$$g_a^*(h) = T_g, \qquad g_t^*(h) = T_g + (t - a)\Delta \quad \text{for } a < t \le b, \tag{70}$$

where $T_g \sim \text{Exp}(\lambda_g)$ is a random variable giving the employment spell-elapsed-time at wave a. Under $m_a = 0$, $g_a = T_g$ (observed).

A wave with $s_t = 0$ does not contribute to the tenure emission (the survey records timegap, not tenure) but its latent clock value still propagates the spell.

C.3.4 Per-spell tenure emission (the key innovation)

For each maximal latent E-spell $[a, b]$ under history h, let $\mathcal{O} \subseteq \{a, a+1, \ldots, b\}$ denote the wave indices with $s_t = 1$ (meaning $\mathcal{O}$ gives the set of observed tenure), and let $t_1 < t_2 < \cdots < t_K$ be the elements of $\mathcal{O}$ with $K = |\mathcal{O}|$. Each observed wave has a latent contamination indicator $m_t \in \{0, 1\}$, drawn independently with $P(m_t = 1) = \varepsilon$. Write $\mathbf{m} = (m_{t_1}, \ldots, m_{t_K}) \in \{0, 1\}^K$ for a contamination pattern, $\mathcal{C}(\mathbf{m}) = \{t \in \mathcal{O} : m_t = 0\}$ for the set of *clean* waves, and $\mathcal{D}(\mathbf{m}) = \{t \in \mathcal{O} : m_t = 1\}$ for the *contaminated* waves.

The joint tenure emission for the spell is the exact mixture over all 2^K contamination patterns:

$$\ell_g^{\text{spell}}\big(\{g_t\}_{t\in\mathcal{O}} \mid h, \Theta\big) \;=\; \sum_{\mathbf{m}\in\{0,1\}^K} \left[\prod_{t\in\mathcal{O}} \varepsilon^{m_t}(1-\varepsilon)^{1-m_t}\right] f\big(\{g_t\} \mid \mathbf{m}, mathbfh\big), \tag{71}$$

where $f(\{g_t\} \mid \mathbf{m}, mathbfh)$ is the conditional density of the observed tenures given the contamination pattern, $\mathbf{m}$, and true employment history, $\mathbf{h}$, obtained by marginalising the latent spell-start elapsed time $T_g \sim \text{Exp}(\lambda_g)$:

- If at least one wave is clean ($\mathcal{C}(\mathbf{m}) \neq \emptyset$), let $t_c = \min(\mathcal{C})$ be the earliest clean wave. The clean report at t_c pins $T_g = g_{t_c} - (t_c - a)\Delta$, giving

  $$f \;=\; \lambda_g\, e^{-\lambda_g\left(g_{t_c} - (t_c-a)\Delta\right)} \cdot \prod_{\substack{t\in\mathcal{C}\\ t>t_c}} \mathcal{I}\big\{g_t = g_{t_c} + (t - t_c)\Delta\big\} \cdot \prod_{t\in\mathcal{D}} \lambda_g\, e^{-\lambda_g g_t} \tag{72}$$

  for $g_{t_c} \geq (t_c - a)\Delta$, and zero otherwise. The first factor is the $\text{Exp}(\lambda_g)$ density of T_g; the indicator product enforces clock-consistency among the remaining clean waves; the final product gives the independent $\text{Exp}(\lambda_g)$ density of each contaminated wave.

- If all waves are contaminated ($\mathcal{C}(\mathbf{m}) = \emptyset$), T_g integrates to 1 and

  $$f \;=\; \prod_{t\in\mathcal{O}} \lambda_g\, e^{-\lambda_g g_t}.$$

Singletons ($K = 1$). When $K = 1$ and $t_1 = a$ (i.e. the observed wave is the spell start), both contamination patterns yield $\lambda_g e^{-\lambda_g g_{t_1}}$, so ε drops out and the emission reduces to a single $\mathrm{Exp}(\lambda_g)$ evaluation. When $t_1 > a$ (the spell start is unobserved), the clean branch is a *shifted* $\mathrm{Exp}(\lambda_g)$ density with support $g_{t_1} \geq (t_1 - a)\Delta$, while the contaminated branch is the unshifted $\mathrm{Exp}(\lambda_g)$ on $[0, \infty)$; the two differ by the factor $e^{\lambda_g (t_1 - a)\Delta}$, and ε remains active.

Pairs ($K = 2$): the identification-critical case. For state pattern $s = (1, 0, 1)$ under $h = (1, 1, 1)$: spell $[1, 3]$, $\mathcal{O} = \{1, 3\}$, $a = 1$, $K = 2$. The contamination indicators (m_1, m_3) take four possible values, which are mutually exclusive and exhaustive. By the law of total probability, the joint emission marginalises over them: each line below is one term, prior probability of the pattern $\times$ conditional density of (g_1, g_3) under that pattern, with the conditional density obtained by integrating out the latent spell-start time $T_g \sim \mathrm{Exp}(\lambda_g)$:

$$
\begin{aligned}
\ell_g^{\text{spell}} \;=\; & (1-\varepsilon)^2\, \lambda_g\, e^{-\lambda_g g_1}\, \mathcal{I}\{g_3 = g_1 + 2\Delta\} && (m_1{=}0,\ m_3{=}0)\\
& + (1-\varepsilon)\, \varepsilon\, \lambda_g^2\, e^{-\lambda_g (g_1 + g_3)} && (m_1{=}0,\ m_3{=}1)\\
& + \varepsilon\, (1-\varepsilon)\, \lambda_g^2\, e^{-\lambda_g (g_1 + g_3 - 2\Delta)} && (m_1{=}1,\ m_3{=}0)\\
& + \varepsilon^2\, \lambda_g^2\, e^{-\lambda_g (g_1 + g_3)} && (m_1{=}1,\ m_3{=}1)
\end{aligned}
\tag{73}
$$

where $2\Delta = 0.5$ years. Line 1: both reports clean — g_1 pins T_g, g_3 must equal $g_1 + 2\Delta$ (point mass). Line 2: wave 1 clean (pins $T_g = g_1$), wave 3 an independent Exp draw. Line 3: wave 1 contaminated, wave 3 clean — T_g is pinned by g_3: $T_g = g_3 - 2\Delta$, producing a *shifted* $\mathrm{Exp}(\lambda_g)$ density for T_g (valid for $g_3 \geq 2\Delta$). Line 4: both contaminated, T_g integrates out.

Why this is the engine of identification. For state pattern $s = (1, 0, 1)$, the two contending latent histories yield different ℓ_g^{spell}:

- Under $h = (1, 1, 1)$ (one E-spell of length 3, with miscl. at $t = 2$): $\mathcal{O} = \{1, 3\}$, $K = 2$. Equation (73) contributes a point mass at $\{g_3 = g_1 + 2\Delta\}$ (line 1), plus three continuous terms (lines 2–4).

- Under $h = (1, 0, 1)$ (two separate singleton E-spells): each spell has $K = 1$ with $t_1 = a$. The emission reduces to $\lambda_g e^{-\lambda_g g_t}$ for each, with no point mass and no dependence on ε.

When $g_3 = g_1 + 2\Delta$ (about 24.8% of $s = (1, 0, 1)$ observations in the QLFS), the $(1, 1, 1)$ hypothesis fires the point mass and dominates, pushing the posterior weight toward $(1, 1, 1)$ and increasing the implied π. When g_3 is inconsistent with $g_1 + 2\Delta$, line 1 of (73) contributes zero, but the three continuous terms (lines 2–4) remain active and still depend on ε. The assignment is then governed primarily by the Markov prior, with secondary adjustment from the relative magnitudes of the continuous contamination terms under $(1, 1, 1)$ versus the two-singleton product $\lambda_g^2 e^{-\lambda_g(g_1+g_3)}$ under $(1, 0, 1)$.

Triples ($K = 3$). For $s = (1, 1, 1)$ under $h = (1, 1, 1)$: $\mathcal{O} = \{1, 2, 3\}$, $a = 1$, $K = 3$. Equation (71) sums over $2^3 = 8$ contamination patterns following the same structure: the all-clean pattern contributes a point mass at $\{g_2 = g_1 + \Delta,\ g_3 = g_1 + 2\Delta\}$; mixed patterns contribute shifted or unshifted Exp densities depending on which wave anchors T_g; the all-contaminated pattern treats all three tenures as independent Exp draws. This case primarily identifies ε from the empirical fraction of $(1, 1, 1)$ panels whose tenure increments are exactly $+\Delta$ per wave.

Posterior contamination probability. For a spell with $K \geq 2$, let τ_h^{spell} denote the posterior probability that *at least one* of the K observed waves in the spell is contaminated, given the observed tenures and the latent history h. By Bayes' rule,

$$\tau_h^{\text{spell}} \;=\; P\big(\exists\, t \in \mathcal{O} : m_t = 1 \,\big|\, \{g_t\}, h, \Theta\big) \;=\; 1 \;-\; \frac{(1-\varepsilon)^K\, f_{\mathbf{0}}}{\ell_g^{\text{spell}}}, \tag{74}$$

where $f_{\mathbf{0}} = f(\{g_t\} \mid \mathbf{m} = \mathbf{0}, h)$ is the all-clean conditional density (e.g. the first line of (73) in the $K = 2$ case, including the clock-consistency indicator), $(1-\varepsilon)^K$ is the prior probability that every wave is clean, and ℓ_g^{spell} from (71) is the marginal likelihood summing over all 2^K contamination patterns. The numerator $(1-\varepsilon)^K f_{\mathbf{0}}$ is therefore the joint probability of the data and the all-clean event; dividing by ℓ_g^{spell} gives the posterior of the all-clean event, and one minus that is the posterior of its complement.

Two limits are useful. When the data exactly satisfy the clock relations required by the all-clean branch (e.g., $g_3 = g_1 + 2\Delta$ for $K = 2$), the all-clean indicator fires and $f_{\mathbf{0}}$ is finite, while the contaminated branches contribute only continuous density at that point. In the limit where the point mass dominates (formally, when the dominating measure assigns positive weight only to the all-clean event at clock-consistent observations), $\tau \to 0$: the spell is judged clean with certainty. Conversely, when the data violate every clock relation, $f_{\mathbf{0}} = 0$ and $\tau = 1$: the spell must contain at least one contaminated wave. Between these extremes τ interpolates smoothly, and is the quantity that *aggregates* across spells in the M-step (83) to deliver $\hat{\varepsilon}$.

C.3.5 N-spell timegap emission

Nonemployment timegap is observed as a category $k \in \{1, \ldots, 7\}$. We retain the interval-censored $\text{Exp}(\lambda_d)$ emission from the base model:

$$P(D \in [a_k, b_k) \mid \lambda_d) = e^{-\lambda_d a_k} - e^{-\lambda_d b_k}. \tag{75}$$

The discrete category structure already provides robustness against small reporting errors (since within-bin variation does not enter the likelihood). We do not introduce an analogue of ε for timegap in this specification. (This is an explicit design choice; see Section C.6.)

C.3.6 Per-wave emissions for non-spell-pair cases

For waves not falling under the per-spell emission rule (e.g., $h_t = 0$, or $h_t = 1$ with $s_t = 0$ so tenure is unobserved), the emissions follow:

- **Misclassified-as-employed** ($h_t = 0$, $s_t = 1$): the observed g_t is drawn from the population marginal of employed tenure, $g_t \sim \text{Exp}(\lambda_g)$. The wave is treated as a singleton E-segment for emission purposes ($K = 1$, no point mass).

- **Matched/misclassified nonemployed** ($s_t = 0$): timegap emission (75) on observed k_t.

C.3.7 State misclassification

(unchanged from the base non-tenure model)

$$q_{ab}(\pi) = \pi^{\mathcal{I}\{a \neq b\}} \, (1 - \pi)^{\mathcal{I}\{a=b\}}. \tag{76}$$

C.3.8 Observed-data likelihood

For individual i, let $y_i = \big(\{s_{it}\}_{t=1}^{3}, \, \{g_{it}\}_{t:s_{it}=1}, \, \{k_{it}\}_{t:s_{it}=0}\big)$ collect all observed quantities: the three reported employment states, the reported tenures at waves where $s_{it} = 1$, and the reported nonemployment-timegap categories $k_{it} \in \{1, \ldots, 7\}$ at waves where $s_{it} = 0$. Given individual weight w_i, the per-history conditional likelihood is

$$p(y_i \mid h; \Theta) = \underbrace{\left[\prod_{t=1}^{3} q_{s_{it},h_t}(\pi)\right]}_{\substack{\text{state}\\\text{misclassification}}} \cdot \underbrace{\left[\prod_{[a,b]\in\mathcal{S}_g(h)} \ell_g^{\text{spell}}(\cdot)\right]}_{\substack{\text{tenure spell}\\\text{emissions (E-spells)}}} \cdot \underbrace{\left[\prod_{t:h_t=0,\, s_{it}=0} P(k_{it} \mid \lambda_d)\right]}_{\substack{\text{timegap: matched}\\\text{nonemployed}\\(h_t=0,\, s_t=0)}} \cdot \underbrace{\left[\prod_{t:h_t=1,\, s_{it}=0} P(k_{it} \mid \lambda_d)\right]}_{\substack{\text{timegap: misclassified-}\\\text{as-nonemployed}\\(h_t=1,\, s_t=0)}}, \tag{77}$$

where $\mathcal{S}_g(h)$ is the set of maximal E-spells in h, and $P(k_{it} \mid \lambda_d)$ is the interval-censored $\text{Exp}(\lambda_d)$ probability (75) assigned to category k_{it}. The first factor scores how well the reported state sequence matches the latent history h; the second gives the joint density of all observed tenures within each E-spell of h (using the point-mass + Exp mixture from (71)); the third and fourth factors score the timegap reports at non-employment waves — separated only because the two cases generate k_{it} from distinct latent durations (a true N-spell duration when $h_t = 0$, versus a fabricated/recall duration when $h_t = 1$ but the respondent reports nonemployment), although both currently use the same $\mathrm{Exp}(\lambda_d)$ emission. The observed-data log-likelihood is

$$\ell(\Theta) = \sum_{i=1}^{N} w_i \log \left[\sum_{h \in \mathcal{H}} p(H_i = h \mid \alpha, \theta_0, \theta_1) \cdot p(y_i \mid h; \Theta) \right]. \tag{78}$$

C.4 EM algorithm

C.4.1 Complete-data formulation

Introduce indicators $z_{ih} = \mathcal{I}\{H_i = h\}$ and per-spell contamination pattern vectors $\mathbf{m}_{i,[a,b],h} \in \{0,1\}^K$ for each maximal E-spell of length $K \geq 1$ under history h. The complete-data log-likelihood factors into independent blocks for the Markov prior, state misclassification, spell contamination, and the spell tenure / timegap emissions.

C.4.2 E-step

Responsibilities.

$$\gamma_{ih} = \frac{p(h \mid \Theta^{\text{old}}) \cdot p(y_i \mid h; \Theta^{\text{old}})}{\sum_{h'} p(h' \mid \Theta^{\text{old}}) \cdot p(y_i \mid h'; \Theta^{\text{old}})}. \tag{79}$$

Posterior contamination probabilities. For each *eps-informative* E-spell under history h — i.e. each maximal E-spell $[a,b]$ with either $K \geq 2$ observed waves, or $K = 1$ with $t_1 > a$ ($K = 1$ spells with $t_1 = a$ are not eps-informative because both contamination patterns yield the same density) — the per-wave posterior contamination probability is

$$\nu_{it,h,[a,b]} \;=\; P(m_t = 1 \mid y_i, h, \Theta^{\text{old}}) \;=\; \frac{\sum_{\mathbf{m}:\, m_t=1} P(\mathbf{m})\, f(\{g_{it}\} \mid \mathbf{m}, h)}{\ell_g^{\text{spell}}(\{g_{it}\} \mid h, \Theta^{\text{old}})}, \tag{80}$$

where the sum is over all contamination patterns with $m_t = 1$, and the denominator is the full spell emission (71). The spell-level posterior probability that at least one wave is contaminated is $\tau_{ih,[a,b]} = 1 - (1-\varepsilon)^K f_{\mathbf{0}}/\ell_g^{\text{spell}}$, as in (74).

C.4.3 M-step

The M-step decomposes into independent blocks because the complete-data log-likelihood is additively separable. We summarise each.

Markov prior block $(\alpha, \theta_0, \theta_1)$: unchanged from the base free specification.

$$\hat{\alpha} = \frac{C_1}{C_0 + C_1}, \qquad \hat{\theta}_1 = \frac{T_{11}}{D_1}, \qquad \hat{\theta}_0 = \frac{T_{01}}{D_0}. \tag{81}$$

Misclassification (π): unchanged.

$$\hat{\pi} = \frac{M}{3W}, \qquad W = \sum_i w_i. \tag{82}$$

Tenure contamination (ε): closed form. The standard EM update for a per-wave Bernoulli mixture weight uses the posterior contamination probabilities $\nu_{it,h}$:

$$\hat{\varepsilon} \;=\; \frac{\sum_{i,h,[a,b]:\,\text{eps-inf}} w_i \gamma_{ih} \sum_{t\in\mathcal{O}} \nu_{it,h,[a,b]}}{\sum_{i,h,[a,b]:\,\text{eps-inf}} w_i \gamma_{ih}\, K_{ih,[a,b]}} \tag{83}$$

where the sums run over all eps-informative spells: those with $K \geq 2$, or $K = 1$ with $t_1 > a$ (the observed wave is after the spell start, so the clean and contaminated branches differ by a shift of $(t_1 - a)\Delta$). $K_{ih,[a,b]}$ is the number of waves with $s = 1$ in the spell. The numerator counts the posterior-expected number of contaminated wave-tenure reports; the denominator counts all wave-tenure reports in eps-informative spells. Spells with $K = 0$, or $K = 1$ with $t_1 = a$, contribute no information about ε and are excluded from both sums.

Spell-length rate (λ_g): closed form (Exponential MLE),

$$\hat{\lambda}_g = \frac{\sum_{i,h,[a,b],t\in\mathcal{O}} w_i \gamma_{ih} \cdot \xi_{it}}{\sum_{i,h,[a,b],t\in\mathcal{O}} w_i \gamma_{ih} \cdot \xi_{it} \cdot g_{it}^{\text{eff}}}, \tag{84}$$

where ξ_{it} is the Exp-weight assigned to observation t by the spell-emission decomposition (1 for the leading observation of a clean spell-pair, 1 for each observation of a contaminated spell-pair, 1 for singletons), and g_{it}^{eff} is the effective tenure (the observed value, except for the leading wave of a clean spell-pair where the relation $g_t = g_a + (t - a)\Delta$ is enforced and only g_a contributes). This is an Exponential-MLE inverse-mean.

Nonemployment rate (λ_d): unchanged from the base EM tenure model. Brent solver on the interval-censored Exp likelihood. The full timegap emission machinery – marginal interval probability (75), conditional transition probability for nonemployment continuations with both flanks observed, and the start emission for fresh nonemployment spells – is reused verbatim from the base / ρ implementations. This preserves apples-to-apples comparability of $\hat{\lambda}_d$ across specifications and concentrates the structural change of Spec I in the tenure block.

C.5 Identification

C.5.1 Block separation of the complete-data log-likelihood

The complete-data log-likelihood decomposes into four independent blocks:

1. Markov prior $\rightarrow$ identifies $(\alpha, \theta_0, \theta_1)$.
2. State misclassification $\rightarrow$ identifies π.
3. Tenure spell-pair contamination $\rightarrow$ identifies ε.
4. Tenure spell-length and N-spell rate $\rightarrow$ identifies (λ_g, λ_d).

C.5.2 Identification of ε

ε is identified from the empirical fraction of E-spell-pairs that satisfy the exact clock-consistency relation in the data. For two consecutive observed waves with

$h_{t-1} = h_t = 1$ and $s_{t-1} = s_t = 1$, the indicator $\mathcal{I}\{g_t - g_{t-1} = \Delta\}$ is Bernoulli with mean $1 - \varepsilon$. Empirically, 63.2% of such pairs satisfy this relation in the QLFS, giving a method-of-moments estimate $\varepsilon \approx 0.37$. For 2-wave bridges ($s = (1, 0, 1)$ with same-spell), the indicator $\mathcal{I}\{g_3 - g_1 = 2\Delta\}$ has mean $(1-\varepsilon)^2$; the empirical 24.8% gives $\varepsilon \approx 0.50$. The EM estimate combines both via the responsibility-weighted average and is consistent under standard regularity.

C.5.3 Identification of π (the corroboration channel)

π is identified from state-Markov violations weighted by tenure flank consistency. Specifically, the M-step formula $\hat{\pi} = M/(3W)$ applies the standard counts to the responsibilities γ_{ih}, but γ_{ih} itself is now a function of the spell-pair tenure emissions. The implications:

- State pattern $s = (1, 0, 1)$ with $g_3 \approx g_1 + 0.5$ yr (24.8% of cases): γ pushed toward $h = (1, 1, 1)$ by the spell-pair point mass. These contribute $\pi \cdot (1-\pi)^2$ to the misclassification count M.
- State pattern $s = (1, 0, 1)$ with g_3 fresh-start-consistent (≤ 3 months, 25% of cases): no point mass under $(1, 1, 1)$; both hypotheses yield single-Exp evaluations. γ split by Markov prior alone, with neither hypothesis dominating.
- State pattern $s = (0, 1, 0)$ with timegap-consistent flanks (e.g., category 1 in both flanks): γ pushed toward $h = (0, 0, 0)$ by the N-spell timegap consistency.

The result is that $\hat{\pi}$ is identified from *both* state-Markov violations (the simple-model channel) *and* tenure flank consistency (the new corroboration channel). When both channels agree, $\hat{\pi}$ is sharper and slightly higher than the simple-model estimate, because tenure flank evidence has reclassified a subset of ($s_{t-1} = 0, s_t = 1$) state transitions from genuine to misclassified.

Specification should yield π modestly above the simple-model estimate of 2.5%-3%. Numerical simulation (a single M-step, holding all else equal, on the QLFS sample) suggests $\hat{\pi}$ in the range 3–5%, depending on $\hat{\varepsilon}$. In this case, tenure data corroborates and refines the simple-model misclassification estimate.

C.5.4 How ε affects the responsibilities

The contamination parameter ε does not enter the Markov prior $p(h \mid \alpha, \theta_0, \theta_1)$; it enters the responsibilities γ_{ih} exclusively through the spell-pair tenure emission ℓ_g^{spell}, which is part of the per-history likelihood $p(y_i \mid h; \Theta)$. Its effect is therefore entirely through changing how well each history h explains the observed tenure data.

Mechanism. Consider the state pattern $s = (1, 0, 1)$. Two competing histories dominate the posterior:

- $h = (1, 1, 1)$: one continuous E-spell, $K = 2$, spell-pair emission ℓ_g^{spell} from (73) — depends on ε.
- $h = (1, 0, 1)$: two singleton E-spells, each with $K = 1$ and $t_1 = a$. Both contamination patterns yield the same $\text{Exp}(\lambda_g)$ density, so ε drops out entirely.

Because ε appears in the numerator (via the $(1, 1, 1)$ spell-pair emission) but not in the relevant terms of competing hypotheses like $(1, 0, 1)$, the level of ε directly controls how much the spell-pair evidence shifts the posterior toward or away from continuous employment.

Clock-consistent observations ($g_3 = g_1 + 2\Delta$). When the tenure flanks satisfy the exact clock relation, the all-clean term $(1 - \varepsilon)^2 \lambda_g e^{-\lambda_g g_1} \, \delta(g_3 - g_1 - 2\Delta)$ fires its point mass and dominates the emission. A higher ε reduces the weight $(1 - \varepsilon)^2$ on this point mass, marginally weakening the evidence for $h = (1, 1, 1)$. However, even at $\varepsilon = 0.25$, the factor $(1 - \varepsilon)^2 = 0.56$ times a point mass still dominates the continuous density from the competing $(1, 0, 1)$ hypothesis, so $\gamma_{i,(1,1,1)}$

remains large. The net effect: higher ε slightly dilutes, but does not eliminate, the point-mass evidence for continuous employment.

Clock-inconsistent observations ($g_3 \neq g_1 + 2\Delta$). When the flanks violate the clock relation, the all-clean point-mass term is exactly zero. The three remaining contamination terms (lines 2–4 of (73)) are continuous and remain active. A higher ε increases the weight on these contamination patterns, making the $(1,1,1)$ hypothesis *more competitive* with $(1,0,1)$, because higher ε makes it more plausible that the individual was continuously employed but one or both tenure reports were simply contaminated. The posterior $\gamma_{i,(1,1,1)}$ therefore *increases* with ε for clock-inconsistent observations, shifting more weight toward continuous employment and increasing the implied contribution to misclassification.

Discrimination power. Combining both regimes, ε controls the *discrimination power* of the spell-pair emission:

- **Low** ε: spell-pairs are highly discriminating. Clock-consistent data produce very high $\gamma_{i,(1,1,1)}$; clock-inconsistent data produce very low $\gamma_{i,(1,1,1)}$. Tenure evidence strongly informs π.
- **High** ε: spell-pairs are less discriminating. Even clock-inconsistent tenure is plausible under $h = (1,1,1)$ (it is merely contaminated). The Markov prior plays a larger role in resolving γ, and tenure's corroborative power for π is diluted.

This is why ε and π are jointly identified but play distinct roles in the posterior: ε governs how *informative* tenure is about latent employment continuity, while π governs how much misclassification there actually is. The EM finds the (ε, π) pair that best explains both the fraction of clock-consistent observations and the state-pattern frequencies simultaneously.

C.5.5 Identification of λ_g

λ_g is identified from the empirical mean of (a) wave-1 employed tenure ($\sim$ 189k observations), (b) misclassified-as-employed tenures (small fraction), (c) the leading observation of clean spell-pairs, and (d) all observations within contaminated spell-pairs. As a single-parameter Exponential MLE, identification is immediate from the inverse-mean.

C.6 Comparison with base EM tenure model

Feature	Base model	Contamination model
Spell-length distribution	$\mathrm{EMG}(\lambda_g, \sigma_g^2)$	$\mathrm{Exp}(\lambda_g)$
Tenure noise	$N(0, 2\sigma_g^2)$	point mass $\delta(\Delta = \Delta)$ or $\mathrm{Exp}(\lambda_g)$ through ε
Tenure-flank $\to \pi$ ID	no	yes (spell-pair)
σ_g^2 collapse issue	yes	not present (no σ)
Free parameters	7	7

C.6.1 Why no ε_d for timegap

The QLFS timegap variable is a 7-category interval-censored measurement. Within-bin variation does not enter the likelihood, providing implicit robustness against small reporting errors. The deterministic clock for nonemployment durations advances 3 months per wave; for a 24-month timegap at wave 1, the clock predicts category 5 ([12,36) months) at all three waves. Bin-crossing happens primarily at the boundaries of categories (e.g., category 1 to 2 at 3 months). Adding an ε_d parameter would be redundant with the categorical robustness already in place and weakens identification of λ_d. We therefore omit it. (A robustness check version with ε_d can be added if a referee requests it.)

C.6.2 Why no ρ (full mismatch)

In the conversation that produced this spec, the user is interested in also reporting a *unified mismatch* model where ρ governs joint state-and-tenure mismatch (Story B). This is preserved in the existing ρ implementation (see `EM tenure rho.tex`) and run in parallel with Spec I. The two specifications target different stories and are complementary:

- Spec I (ε only): tenure errors generate inconsistent reports while state is correct. π is corroborated and refined by tenure flanks.
- ρ model: a fraction of observations is jointly random (state-and-tenure unrelated to true trajectory). π is identified from non-mismatched observations only.

C.7 Nesting and tests

Nesting. The base specification is recovered by setting $\varepsilon = 0$ and replacing the spell-pair emission with an EMG-and-Gaussian-increment evaluation. Spec I does *not* formally nest the base model because the distributional families differ (point-mass-Exp vs. EMG-Gaussian); however the comparison can be made via non-nested model selection (BIC, Vuong test).

LR test of $\varepsilon = 0$. Within Spec I, the restriction $\varepsilon = 0$ is testable. Under $H_0 : \varepsilon = 0$ the spell-pair emission collapses to deterministic clock; observations with non-exact increments contribute zero likelihood. The unrestricted Spec I dominates trivially given the data.

LR test of CTMC-link. The linked specification $\lambda_g = -\log(\theta_1)/\Delta$ can be applied to Spec I exactly as in the base model. The LR test is χ^2_2 under the null.

C.8 Preliminary results

Here we present preliminary evidence on the model presented above. First, some descriprive evidence from the QLFS data are given. Second, the estimation of the tenure contamination model is given.

C.8.1 Descriptive evidence from the QLFS

We now quantify the empirical regularities noted in Section 1 more systematically. Table 14 reports diagnostics computed from the three-wave QLFS panel.

Table 14: Descriptive duration statistics from the QLFS

Diagnostic	N	Share
E→E continuation increments (consecutive employed waves)		
Exact +3-month increment ($\Delta g = \Delta$)	343 850	63.2%
Deviation > 12 months from $+\Delta$		18.7%
State pattern $s = (1, 0, 1)$*: tenure-flank classification*		
$g_3 \approx g_1 + 2\Delta$	5 997	27.4%
$g_3 \leq 3$ months (fresh-start consistent)		18.3%
Within-panel fresh starts ($s_{t-1} = 0$, $s_t = 1$)		
Tenure > 12 months at entry wave	35 389	36.7%
State pattern $s = (0, 1, 0)$*: timegap-flank consistency*		
$k_1 = k_3$ (same timegap category at both flanks)	6 445	63.7%
Method-of-moments $\hat{\varepsilon}$		
From adjacent E→E pairs: $\hat{\varepsilon} = 1 - 0.632$		0.368
From $(1, 0, 1)$ bridges: $\hat{\varepsilon} = 1 - \sqrt{0.274}$		0.476

Notes. N is the number of wave-pairs or panels contributing to each diagnostic. Shares are unweighted.

Three features stand out. First, 63.2% of consecutive employed-wave pairs report the exact clock-consistent increment of +3 months, while 18.7% deviate by more than a year. This bimodal pattern—a point mass at the deterministic clock value plus a heavy-tailed residual—directly motivates the point-mass + Exp contamination mixture. A simple method-of-moments estimator gives $\hat{\varepsilon} \approx 0.37$ from adjacent pairs.

Second, among within-panel fresh starts ($s_{t-1} = 0, s_t = 1$), 36.7% report tenure exceeding 12 months. Since a genuine job entry cannot have more than $\Delta = 3$ months of tenure at the survey wave, these observations imply misclassification of employment status at $t-1$ and provide a model-free lower bound on the misclassification rate among observed $(0 \to 1)$ transitions.

Third, for the state pattern $s = (1, 0, 1)$, 27.4% of panels have tenure flanks consistent with continuous employment ($g_3 \approx g_1 + 2\Delta$), providing direct evidence for the corroboration channel of Section C.5. An additional 18.3% have short tenure at wave 3 ($g_3 \leq 3$ months), consistent with genuine exit and re-entry. The remaining ∼54% are ambiguous or show year-rounding artefacts.

C.8.2 Estimation results

Table 15 reports parameter estimates from the contamination model under specifications alongside the baseline simple models that use employment states alone. The preferred specification (stationary, CTMC-linked) is in the rightmost column.

Table 15: Parameter estimates: baseline models and contamination model

	Simple (no tenure)		Contamination model	
	No ME	Symm. ME	Linked	**Stat–Linked**
Entry rate θ_0 (%)	8.34	2.90	4.08	**4.14**
Exit rate $1-\theta_1$ (%)	8.36	2.91	4.26	**4.26**
Miscl. rate π (%)	—	2.98	10.21	**10.02**
Contam. rate ε (%)	—	—	24.79	**24.69**
λ_g	—	—	0.174	**0.174**
λ_d	—	—	0.167	**0.169**
Mean E spell (months)	—	—	68.9	**68.9**
Mean N spell (months)	—	—	72.0	**71.0**
Init. empl. prob. α	—	—	0.550	**0.493**
CTMC link	—	—	Yes	Yes
Stationarity	—	—	No	Yes

Notes. Simple models estimated by MLE on three-wave employment state sequences ($N = 22{,}779$). Contamination models estimated by EM on three-wave panels with tenure and timegap ($N = 402{,}923$). "Linked" constrains $\lambda_g = -\log\theta_1/\Delta$ and $\lambda_d = -\log(1-\theta_0)/\Delta$. "Stationary" constrains $\alpha = \theta_0/(\theta_0 + 1 - \theta_1)$.

Contamination rate. $\hat{\varepsilon}$ is stable across all variants at $\approx$ 24–25%, below the method-of-moments estimates (36.8% adjacent, 47.6% bridges) in Table 14. The gap arises because the MoM attributes *all* non-exact increments to contamination, whereas the EM accounts for the full mixture of latent histories: some non-exact increments arise from genuine transitions (new spells), not contamination within a continuing spell.

Misclassification rate. The contamination model yields $\hat{\pi} \approx 10\%$ under the linked specifications, substantially above the simple ME model's 2.98%. This reflects the corroboration channel: tenure-flank consistency in the state pattern $s = (1, 0, 1)$ provides direct evidence that some observed $(1 \to 0 \to 1)$ sequences are misclassified continuous employment. The simple model, lacking tenure information, identifies π from the $2^3 = 8$ state-pattern frequencies alone. The descriptive finding that 36.7% of within-panel fresh starts have impossibly long tenure (Table 14) is consistent with a misclassification rate well above the simple model's 3%.

Transition rates. Under the preferred stationary-linked specification, the entry rate is $\hat{\theta}_0 = 4.14\%$ and the exit rate is $1 - \hat{\theta}_1 = 4.26\%$, yielding a steady-state employment rate of $\hat{\alpha} = 49.3\%$. These fall between the naïve no-ME rates ($\sim$8.3%) and the simple ME rates ($\sim$2.9%). The naïve model overstates transitions because observed state changes include misclassifications. The simple ME model understates them because, without tenure information, it assigns too little probability to misclassification and consequently deflates implied true transitions.

C.9 Summary

The contamination model replaces the Gaussian-EMG measurement model with a structurally correct point-mass + Exp contamination mixture and introduces a spell-pair joint emission that allows tenure flanks to identify misclassification. The net result is no σ_g^2 collapse issue, and a direct identification channel for π via tenure flank consistency in same-spell wave pairs.